

THE INVERSE LEIBNIZ PROBLEM: RECONSTRUCTION FIBERS, RIGIDITY, OBSTRUCTIONS, AND DEFORMATION DGLAS

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ABSTRACT. We reconstruct left and right algebra actions from a fixed bilinear map $B : A \times A \rightarrow V$ required to satisfy two Leibniz identities. The weak reconstruction fiber is affine; strict reconstruction adds the quadratic equations for a unital bimodule. Finite examples distinguish global uniqueness, infinitesimal rigidity, and nonreduced tangent directions.

For $A = \mathbb{C}[\epsilon]/(\epsilon^4)$ and $V = \mathbb{C}^3$, the completed local ring is $\mathbb{C}[[u, v]]/(v^3)$. Every normalized contraction of the presentation DGLA has a nonzero quartic operation, whereas a single contraction of the constrained Hochschild DGLA retains the cubic obstruction and has vanishing degree-one operations at arities 4 through 23. A second fixed contraction has a nonzero arity-23 coefficient.

For the preserved low-rail homotopy data, an autonomous triangular subsystem has an all-orders scalar recurrence. We determine its residual modules, complete landing operators, and the nested gcd stratifications of forcing, spectral support, and a specified polynomial lift. Exact cochain and matrix certificates accompany the proofs; two frozen out-of-sample checks provide separate finite evidence. The spectral results are relative to the fixed contraction, not invariants of an unmarked deformation problem.

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1. INTRODUCTION

Classical deformation theory typically starts with an algebraic structure and studies its infinitesimal and formal variations. In the present problem, the direction is partly reversed. We are given a bilinear map

$$B : A \times A \longrightarrow V$$

and ask whether the left and right A -actions on V can be recovered from the requirement that B satisfy Leibniz rules in both variables.

The key typing observation is elementary but decisive: a formula such as

$$B(ab, c) = a B(b, c) + B(a, c) b$$

is meaningless unless the products between A and V have already been defined. Thus “ B is a biderivation” is a property of a triple (B, L, R) , not of B alone. The inverse Leibniz problem asks how much of (L, R) is observable from B .

The results are organized into five groups.

- (i) Reconstruction and rigidity: the weak affine fiber, the strict scheme, and examples distinguishing global ambiguity from infinitesimal rigidity (Sections 3–8).
- (ii) Formal obstruction theory: presentation and constrained Hochschild DGLAs, effective relation spaces, and the cubic germ $\mathbb{C}[[u, v]]/(v^3)$ (Sections 10–12).
- (iii) Homotopy transfer: a complete single-contraction certificate through arity 23, exact feedback matrices, and a contrasting frozen completion (Section 13).
- (iv) All-orders low-rail dynamics: residual modules with ranks $(1, 2, 8)$ and the annihilators $p_3 = E^2 q$, $p_4 = E^3 q$, $p_5 = E^4 q^2$ (Section 15).
- (v) Finite spectral selection: landing operators and the nested invariants $\gcd(m_A, r_d)$, $\gcd(p_4, r_d)$, and $\gcd(p_5, r_d)$ (Sections 16–17).

The richer fixed-contraction spectrum studied here is compared under normalized homotopy tilts in Paper II [12]; its surviving spectral floor is generalized, under explicit finite marked hypotheses, in Paper III [13]. Those comparisons do not make the quintic factor q an unmarked invariant. The present paper supplies the fixed calculation on which that distinction is based.

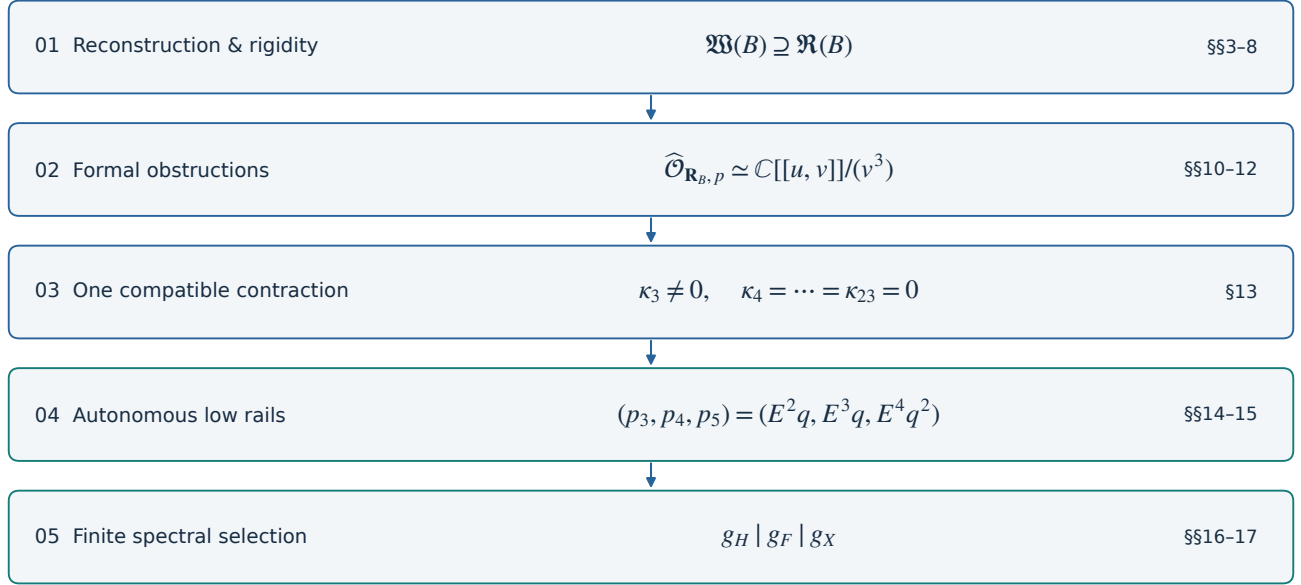
The framework is adjacent to, but directionally different from, universal differential calculi and double-derivation formalisms, where a target bimodule structure is normally fixed in advance. It also follows the standard deformation-theoretic pattern in which tangent vectors are cocycle-like data and higher-order extension is controlled by obstruction classes [1, 2]. Section 1.1 gives a bounded comparison with the neighboring deformation, module, differential, and representation-space frameworks; it is a scope statement rather than a claim of exhaustive novelty.

The overall reading map is shown in Figure 1.

1.1. Position relative to adjacent frameworks. The inverse Leibniz problem should be distinguished from several established deformation theories. The common ground is the use of formal equations, tangent spaces, and higher obstruction classes. The decisive difference is the direction of reconstruction: here A , V , and the bilinear observation B are fixed, while the left and right actions are the unknowns. The reconstruction fiber is therefore a framed action fiber, not the deformation functor of the multiplication on A .

Concretely, $\mu(a \otimes b^{\text{op}}) = L(a)R(b)$ identifies strict action pairs with representations of A^e on the fixed vector space V . The Leibniz identities cut out the fixed- B locus in that representation scheme. Section 10.2 gives the corresponding identification of framed Artin deformations. The reconstruction problem is therefore a constrained representation-deformation problem.

Reading map: from a fixed bilinear map to the spectrum of a fixed contraction.



Fixed B for reconstruction • chosen homotopy for the spectral calculation

FIGURE 1. The five result groups of Paper I. The rows give a reading order, not additional implication claims. The transfer endpoint uses one compatible contraction; the spectral calculation retains the chosen low-rail homotopy.

Accordingly, the contribution claimed in this version is narrow and verifiable: a fixed cubic inverse-Leibniz example, its exact formal germ, the intrinsic higher-transfer and low-rail certificates, and the resulting finite spectral stratifications. The standard deformation and homotopy-transfer machinery is used in the sense of [7, 10, 11]. The contribution assessed here is the fixed reconstruction calculation and its spectral selection, rather than a new general deformation formalism.

Status and limitations. All finite certificates use exact arithmetic. The two complete intrinsic contractions share the saved low-rail assignments but differ at arity 23; neither gives a universal finite stopping bound or an all-orders pure-cubic model. The all-orders theorem concerns only the fixed autonomous low rails. Unsupported historical detector normalizations are excluded from the proof chain; their provenance is recorded once in Appendix A. The scope fixes A, V, B and the displayed homotopy data, without claiming literature-wide novelty.

Notation. In the deformation sections, A is the underlying algebra, $E = \text{End}(V)$, and $S = A^e$. In the spectral sections, $A = -\hbar \text{ad}_\alpha$ is an operator, S is the temporal shift or its polynomial variable, and $E = S + 2$. The cochain e and the polynomial E are distinct. Paper II supplies its own explicit cochain dictionary.

2. TYPED RECONSTRUCTION DATA

Throughout, A is a unital associative algebra over $\mathbb{C}$ and V is a complex vector space. Fix a bilinear map

$$B : A \times A \longrightarrow V.$$

TABLE 1. Bounded literature positioning for the fixed-data inverse problem.

Framework	Object studied or varied	Relation to the present problem
Algebra deformations [1]	The multiplication of an algebra	The multiplication of A is fixed here; the unknowns are two actions on V .
Artin and DGLA deformation theory [2, 10]	A formal functor with tangent and obstruction spaces	The same language organizes our local equations, but the functor is a reconstruction fiber with B held fixed.
Algebra morphisms and diagrams [5, 6]	Structure maps and their compatibility diagrams	The representation $A^e \rightarrow \text{End}(V)$ varies subject to the fixed- B compatibility equations.
Module deformations [8]	Modules or module families over a fixed algebra	We keep A and B fixed and reconstruct the module actions themselves from the observed biderivation.
Universal noncommutative differentials [3]	Universal extensions and differential forms attached to an algebra	Those constructions explain the differential-calculus neighborhood; our exact computations concern a finite-dimensional, fixed- B fiber.
Double brackets and representation spaces [4, 9]	A double operation or a representation scheme of a fixed algebra	The strict fiber is a fixed- B constrained locus in $\text{Rep}_V(A^e)$; no double-Poisson structure or derived enhancement is constructed here.

A candidate left-right action pair consists of linear maps

$$L : A \longrightarrow \text{End}(V), \quad R : A \longrightarrow \text{End}(V).$$

We use the notation

$$a \triangleright v = L(a)v, \quad v \triangleleft a = R(a)v.$$

No multiplicativity is assumed at first.

Definition 2.1 (Weak reconstruction fiber). *The weak reconstruction fiber $\mathfrak{W}(B)$ is the set of pairs (L, R) satisfying*

$$\begin{aligned} (1) \quad & B(ab, c) = L(a)B(b, c) + R(b)B(a, c), \\ (2) \quad & B(a, bc) = R(c)B(a, b) + L(b)B(a, c) \end{aligned}$$

for all $a, b, c \in A$.

Definition 2.2 (Strict reconstruction fiber). *The strict reconstruction fiber $\mathfrak{R}(B)$ is the subset of $\mathfrak{W}(B)$ consisting of pairs satisfying*

$$\begin{aligned} (3) \quad & L(1) = R(1) = I_V, \\ (4) \quad & L(ab) = L(a)L(b), \\ (5) \quad & R(ab) = R(b)R(a), \\ (6) \quad & [L(a), R(b)] = 0 \end{aligned}$$

for all $a, b \in A$.

Thus a point of $\mathfrak{R}(B)$ makes V into a unital A -bimodule for which B is a biderivation relative to the convention in (1)–(2).

2.1. Framed reconstruction and the fixed- B gauge boundary. The fiber above is framed: V and the actual map $B : A \times A \rightarrow V$ are held fixed. The only basis changes that preserve this same inverse problem are the pointwise stabilizers

$$G_B = \{g \in \mathrm{GL}(V) : gB(a, b) = B(a, b) \text{ for all } a, b \in A\}.$$

Proposition 2.3 (Fixed- B gauge action). *The group G_B acts on $\mathfrak{R}(B)$ by*

$$g \cdot (L, R) = (gL(-)g^{-1}, gR(-)g^{-1}).$$

Its Lie algebra is

$$\mathfrak{g}_B = \{Z \in \mathrm{End}(V) : ZB(a, b) = 0 \text{ for all } a, b \in A\}.$$

In particular, if $V = \mathrm{span} B(A, A)$, then $G_B = \{I_V\}$ and $\mathfrak{g}_B = 0$.

Proof. For $g \in G_B$, conjugating each action equation by g preserves the unit, representation, and commuting-action equations. In the weak equations, use $g^{-1}B(a, b) = B(a, b)$ and then apply g to both sides; the fixedness of B gives the original right-hand side. Differentiating the pointwise condition $gB(a, b) = B(a, b)$ at the identity gives the displayed Lie algebra. If $B(A, A)$ spans V , an element of G_B fixes a spanning set pointwise and is therefore the identity; the infinitesimal assertion is identical. $\square$

Thus the four strict points in Section 5, and the later spanning examples, are genuinely distinct points of the fixed- B fiber rather than points identified by a nontrivial fixed- B conjugation. The broader quotient problem begins only when $B(A, A)$ fails to span V or when A and B are allowed to deform.

3. THE LEIBNIZ OBSERVABILITY OPERATOR

Suppose (L, R) and (L', R') lie in $\mathfrak{W}(B)$, and set

$$X = L' - L, \quad Y = R' - R.$$

Subtracting the two weak Leibniz systems gives

$$(7) \quad X(a)B(b, c) + Y(b)B(a, c) = 0,$$

$$(8) \quad Y(c)B(a, b) + X(b)B(a, c) = 0.$$

Definition 3.1 (Leibniz observability operator). *Define*

$$\mathcal{O}_B : \mathrm{Hom}(A, \mathrm{End} V)^2 \longrightarrow \mathrm{Hom}(A^{\otimes 3}, V)^2$$

by

$$\mathcal{O}_B(X, Y)(a, b, c) = \begin{pmatrix} X(a)B(b, c) + Y(b)B(a, c) \\ Y(c)B(a, b) + X(b)B(a, c) \end{pmatrix}.$$

Proposition 3.2 (Affine structure of the weak fiber). *If $\mathfrak{W}(B)$ is nonempty and $(L_0, R_0) \in \mathfrak{W}(B)$, then*

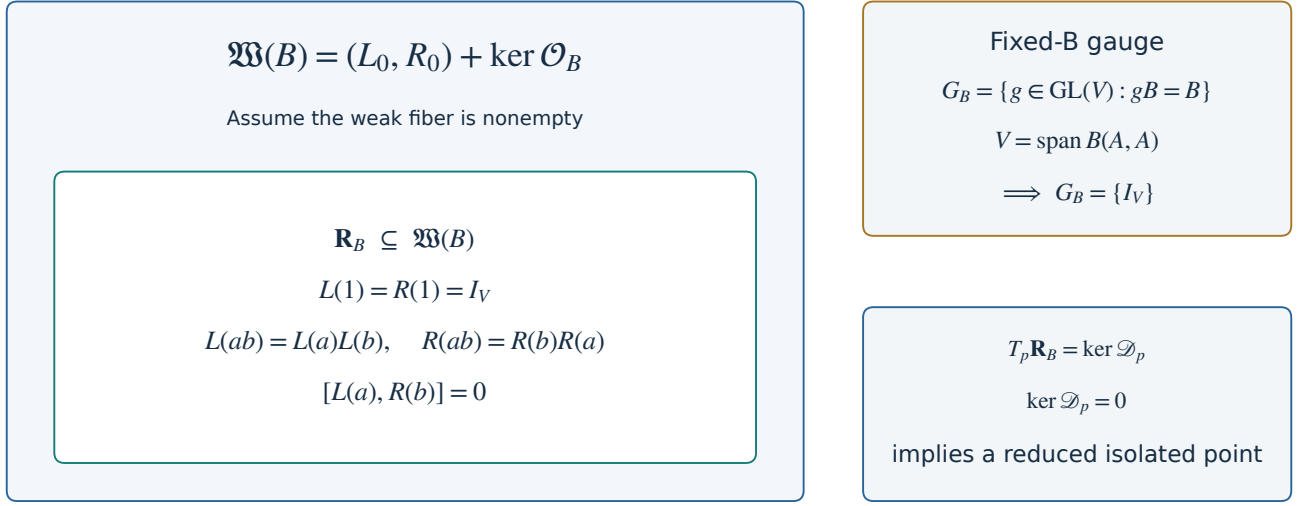
$$\mathfrak{W}(B) = (L_0, R_0) + \ker \mathcal{O}_B.$$

In particular, weak reconstruction is unique if and only if $\ker \mathcal{O}_B = 0$.

Proof. Write $(L, R) = (L_0 + X, R_0 + Y)$. Equations (1)–(2) are affine-linear in (L, R) , and their homogeneous part is exactly (7)–(8). $\square$

The corresponding structures are illustrated in Figure 2.

The weak equations are affine-linear; strictness adds the unital bimodule relations.



Weak uniqueness: $\ker \mathcal{O}_B = 0$. Strict local rigidity and global uniqueness are different questions.

FIGURE 2. Weak and strict reconstruction for fixed B . The nonempty weak fiber is affine, while strictness imposes the unital bimodule equations. The fixed- B stabilizer is trivial when $B(A, A)$ spans V ; strict tangent observability is described by $\ker \mathcal{D}_p$. Nested regions are schematic, not dimension estimates.

4. FINITE-DIMENSIONAL MATRIX REALIZATION

Assume $\dim A = m$ and $\dim V = d$. Choose bases $e_1, \dots, e_m$ of A and $v_1, \dots, v_d$ of V . Write

$$e_i e_j = \sum_{\ell=1}^m C_{ij}^\ell e_\ell, \quad B(e_i, e_j) = \sum_{\beta=1}^d B_{ij}^\beta v_\beta.$$

Let $L_i = L(e_i)$ and $R_i = R(e_i)$. The weak equations become

$$(9) \quad \sum_{\beta} (L_i)_{\alpha\beta} B_{jk}^\beta + \sum_{\beta} (R_j)_{\alpha\beta} B_{ik}^\beta = \sum_{\ell} C_{ij}^\ell B_{\ell k}^\alpha,$$

$$(10) \quad \sum_{\beta} (R_k)_{\alpha\beta} B_{ij}^\beta + \sum_{\beta} (L_j)_{\alpha\beta} B_{ik}^\beta = \sum_{\ell} C_{jk}^\ell B_{i\ell}^\alpha.$$

After ordering all matrix entries, these equations give

$$M_B u = r_B, \quad M_B \in M_{2m^3 d \times 2md^2}(\mathbb{C}).$$

Consequently,

$$\mathfrak{W}(B) \neq \emptyset \iff \text{rank } M_B = \text{rank}[M_B \mid r_B],$$

and, if consistent,

$$\dim \mathfrak{W}(B) = 2md^2 - \text{rank } M_B.$$

The strict equations add the quadratic constraints

$$\begin{aligned} L_i L_j &= \sum_{\ell} C_{ij}^{\ell} L_{\ell}, \\ R_j R_i &= \sum_{\ell} C_{ij}^{\ell} R_{\ell}, \\ L_i R_j &= R_j L_i, \end{aligned}$$

together with the unit equations. Hence in finite dimension the strict reconstruction space is naturally an affine scheme cut out by a linear system and quadratic bimodule equations.

5. A FINITE-DIMENSIONAL NON-IDENTIFIABILITY THEOREM

We now show that the spanning condition $V = \text{span } B(A, A)$ does not force strict uniqueness.

Let

$$q(x) = (x-1)(x-2)(x-3)(x-4)$$

and

$$A = \mathbb{C}[x]/(q(x)).$$

Let $V = \mathbb{C}^2$, $v_0 = (1, 1)^T$, and define

$$L_x = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad R_x = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}.$$

For a polynomial class $p \in A$, set

$$\Delta_{L,R}(p) = (p(L_x) - p(R_x))(L_x - R_x)^{-1}$$

and define

$$B(p, q) = \Delta_{L,R}(p) \Delta_{L,R}(q) v_0.$$

Since all matrices involved are diagonal, the divided-difference identity

$$\Delta_{L,R}(pq) = p(L_x) \Delta_{L,R}(q) + \Delta_{L,R}(p) q(R_x)$$

implies both weak Leibniz identities.

Moreover,

$$B(x, x) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad B(x^2, x) = \begin{pmatrix} 4 \\ 6 \end{pmatrix},$$

which are linearly independent. Thus

$$V = \text{span } B(A, A).$$

The scalar divided difference is symmetric in its two evaluation points. Consequently, independently swapping the two points in either coordinate of V leaves B unchanged. This produces the four strict reconstructions

L_x	R_x
diag(1, 2)	diag(3, 4)
diag(3, 2)	diag(1, 4)
diag(1, 4)	diag(3, 2)
diag(3, 4)	diag(1, 2).

Theorem 5.1 (Global non-identifiability). *There exist finite-dimensional A, V , and $B : A \times A \rightarrow V$ such that*

$$V = \text{span } B(A, A)$$

but $|\mathfrak{R}(B)| = 4$. Hence the spanning condition alone does not imply unique reconstruction.

Proof. The construction above gives at least four strict reconstructions. Exact elimination in the basis $1, x, x^2, x^3$ gives a weak system with 32 unknowns and rank 28. Parameterizing the weak solution by the entries of $R(x^3)$, the strict equations have Groebner basis

$$(R(x^3)_{11} - 1)(R(x^3)_{11} - 27), \quad R(x^3)_{12}, \quad R(x^3)_{21}, \quad (R(x^3)_{22} - 8)(R(x^3)_{22} - 64).$$

Thus there are exactly four strict points, corresponding to the table above. $\square$

6. STRICT TANGENT OBSERVABILITY AND RIGIDITY

Fix $p = (L, R) \in \mathfrak{R}(B)$. Define

$$\begin{aligned} d_L X(a, b) &= X(a)L(b) + L(a)X(b) - X(ab), \\ d_R Y(a, b) &= Y(b)R(a) + R(b)Y(a) - Y(ab), \\ \chi_{L,R}(X, Y)(a, b) &= [X(a), R(b)] + [L(a), Y(b)]. \end{aligned}$$

Definition 6.1 (Strict tangent observability operator). *Define*

$$\mathcal{D}_p(X, Y) = \begin{pmatrix} \mathcal{O}_B(X, Y) \\ X(1) \\ Y(1) \\ d_L X \\ d_R Y \\ \chi_{L,R}(X, Y) \end{pmatrix}.$$

Theorem 6.2 (Strict tangent-space formula). *The Zariski tangent space of the strict reconstruction scheme at p is*

$$T_p \mathbf{R}_B = \ker \mathcal{D}_p.$$

Proof. Substitute $L_\varepsilon = L + \varepsilon X$ and $R_\varepsilon = R + \varepsilon Y$ into the weak Leibniz and bimodule equations over $\mathbb{C}[\varepsilon]/(\varepsilon^2)$. The coefficient of ε is exactly $\mathcal{D}_p(X, Y)$. $\square$

Definition 6.3 (Three notions of rigidity). *A strict reconstruction p is:*

- (i) infinitesimally rigid if $T_p \mathbf{R}_B = 0$;
- (ii) locally rigid if p is isolated in the reduced reconstruction variety;
- (iii) globally unique if $\mathfrak{R}(B) = \{p\}$.

Proposition 6.4 (Infinitesimal rigidity implies local rigidity). *In the finite-dimensional setting, if $T_p \mathbf{R}_B = 0$, then p is a reduced isolated point.*

Proof. The local coordinate ring is Noetherian. Vanishing tangent space gives $\mathfrak{m}_p/\mathfrak{m}_p^2 = 0$. Nakayama's lemma yields $\mathfrak{m}_p = 0$, so the local ring is $\mathbb{C}$. $\square$

For the four points in Theorem 5.1, exact Jacobian computation gives full rank 32 at every point. Therefore each point is infinitesimally and locally rigid, while the reconstruction is not globally unique. This separates local rigidity from global identifiability.

The corresponding structures are illustrated in Figure 3.

7. SECOND-ORDER OBSTRUCTION THEORY

Let $p = (L, R) \in \mathfrak{R}(B)$ and let

$$\xi = (X, Y) \in T_p \mathbf{R}_B.$$

We ask whether ξ extends to a strict reconstruction over $\mathbb{C}[t]/(t^3)$:

$$L_t = L + tX + t^2 X_2, \quad R_t = R + tY + t^2 Y_2.$$

Independent swaps preserve the divided differences defining B .

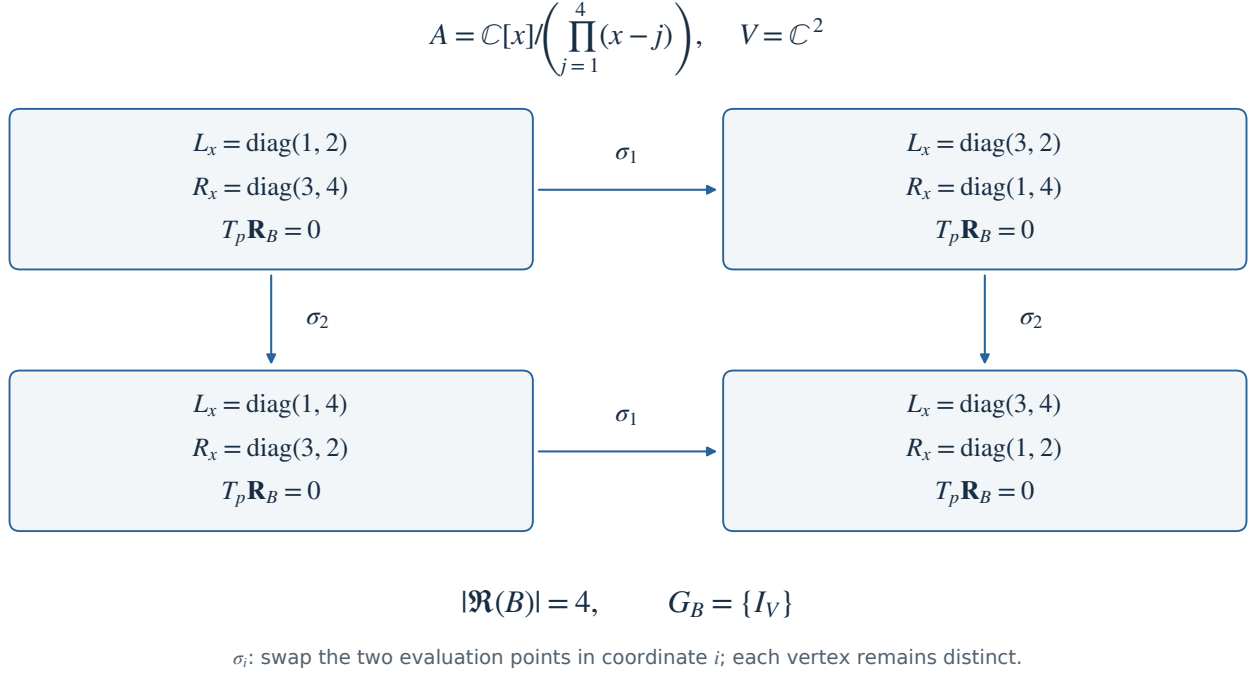


FIGURE 3. The four strict reconstructions of Theorem 5.1. Independent swaps of evaluation points preserve B . Each point is reduced and infinitesimally rigid, but the fixed- B fiber is not globally unique. The arrows are swaps of the displayed constructions, not gauge identifications.

The weak Leibniz equations are linear in (L, R) , so their second-order source term is zero. The quadratic bimodule equations produce

$$(11) \quad d_L X_2(a, b) = -X(a)X(b),$$

$$(12) \quad d_R Y_2(a, b) = -Y(b)Y(a),$$

$$(13) \quad \chi_{L,R}(X_2, Y_2)(a, b) = -[X(a), Y(b)].$$

Together with $\mathcal{O}_B(X_2, Y_2) = 0$ and $X_2(1) = Y_2(1) = 0$, these equations can be written as

$$\mathcal{Q}_p(X_2, Y_2) = -\mathcal{Q}_p(X, Y),$$

where

$$\mathcal{Q}_p(X, Y) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ ((a, b) \mapsto X(a)X(b)) \\ ((a, b) \mapsto Y(b)Y(a)) \\ ((a, b) \mapsto [X(a), Y(b)]) \end{pmatrix}.$$

Definition 7.1 (Second-order obstruction space and map). *Set*

$$\text{Obs}_p(B) = \text{coker } \mathcal{Q}_p$$

and define

$$\text{ob}_{2,p} : T_p \mathbf{R}_B \longrightarrow \text{Obs}_p(B), \quad \text{ob}_{2,p}(X, Y) = [\mathcal{Q}_p(X, Y)].$$

Theorem 7.2 (Primary obstruction criterion). *A strict tangent vector $\xi \in T_p \mathbf{R}_B$ extends to order two if and only if*

$$\text{ob}_{2,p}(\xi) = 0.$$

Proof. An order-two extension exists precisely when the affine equation

$$\mathcal{D}_p(X_2, Y_2) = -\mathcal{Q}_p(\xi)$$

has a solution. This is equivalent to $\mathcal{Q}_p(\xi)$ lying in $\text{im } \mathcal{D}_p$. $\square$

The map $\text{ob}_{2,p}$ is homogeneous quadratic rather than linear. Its polarization is a symmetric bilinear expression whose three nonzero components are

$$\begin{aligned} X(a)X'(b) + X'(a)X(b), \\ Y(b)Y'(a) + Y'(b)Y(a), \\ [X(a), Y'(b)] + [X'(a), Y(b)]. \end{aligned}$$

Section 10 shows that this polarization is already the degree-one bracket of a two-step nilpotent differential graded Lie algebra; no higher L_∞ operations are required for the present framed problem.

8. A NONREDUCED FALSE TANGENT

Let

$$A = \mathbb{C}[\epsilon]/(\epsilon^2), \quad V = \mathbb{C},$$

and define $B : A \times A \rightarrow \mathbb{C}$ by

$$B(\epsilon, \epsilon) = 1$$

and $B = 0$ on the other basis pairs. Then $V = \text{span } B(A, A)$.

A unital one-dimensional action pair is determined by

$$L(\epsilon) = \ell, \quad R(\epsilon) = r.$$

The weak Leibniz equations give

$$\ell + r = 0,$$

while the representation equations give

$$\ell^2 = 0, \quad r^2 = 0.$$

Hence

$$\mathbf{R}_B = \text{Spec } \frac{\mathbb{C}[\ell, r]}{(\ell + r, \ell^2, r^2)} \cong \text{Spec } \frac{\mathbb{C}[\ell]}{(\ell^2)}.$$

The scheme has one geometric point $p = (0, 0)$ but a one-dimensional tangent space

$$T_p \mathbf{R}_B = \mathbb{C}(1, -1).$$

Theorem 8.1 (A first-order direction obstructed at second order). *Every nonzero vector in $T_p \mathbf{R}_B$ is obstructed at second order.*

Proof. For a tangent vector $(x, -x)$, attempt

$$\ell_t = tx + t^2u, \quad r_t = -tx + t^2v.$$

The weak equation requires $u + v = 0$. However,

$$\ell_t^2 = t^2x^2 \pmod{t^3}, \quad r_t^2 = t^2x^2 \pmod{t^3}.$$

Over $\mathbb{C}$, these vanish only when $x = 0$. $\square$

Dual-number example: nilpotent thickness is first-order information.

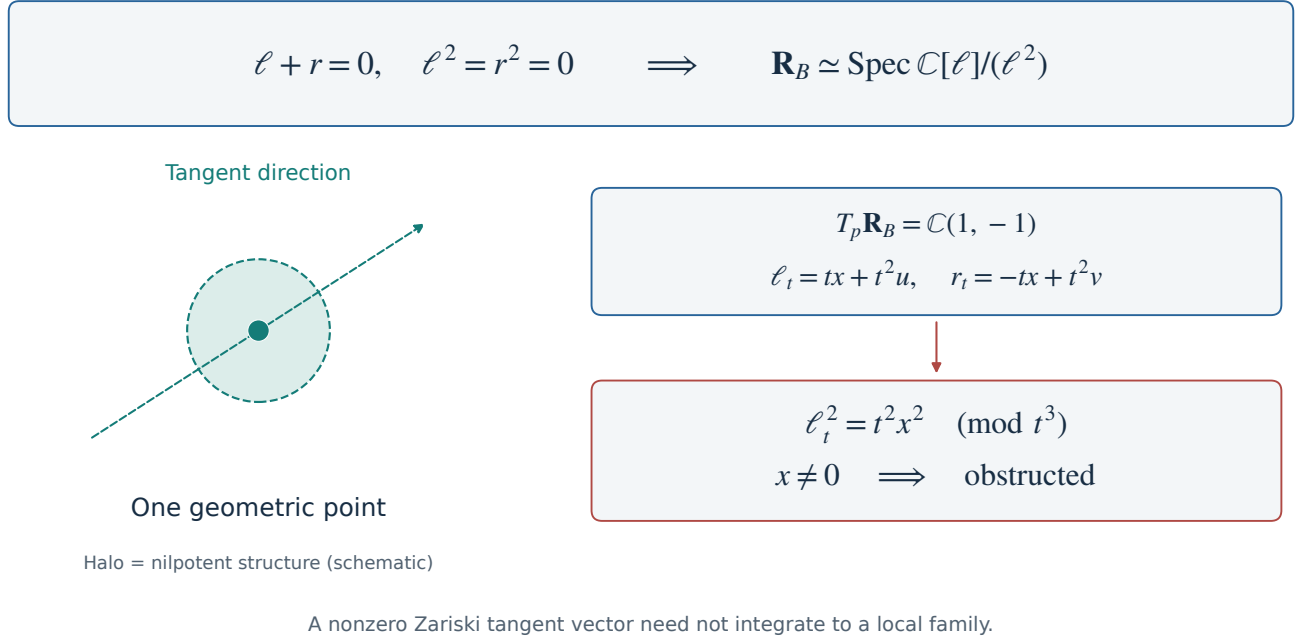


FIGURE 4. The dual-number false tangent of Theorem 8.1. The halo denotes nilpotent structure at one geometric point; the dashed line denotes the Zariski tangent direction, not a family of geometric points. Every nonzero first-order direction is obstructed at second order.

Thus a nonzero Zariski tangent vector need not arise from any genuine local family. It may record only the nilpotent thickness of a nonreduced reconstruction scheme.

The corresponding structures are illustrated in Figure 4.

9. HIGHER-ORDER RECURSION

Suppose

$$L_t = L + \sum_{n \geq 1} t^n X_n, \quad R_t = R + \sum_{n \geq 1} t^n Y_n.$$

At order $n \geq 2$, the extension equation has the form

$$\mathcal{D}_p(X_n, Y_n) = -\mathcal{Q}_n,$$

where the nonzero source terms are

$$\begin{aligned} \mathcal{Q}_n^L(a, b) &= \sum_{i=1}^{n-1} X_i(a) X_{n-i}(b), \\ \mathcal{Q}_n^R(a, b) &= \sum_{i=1}^{n-1} Y_i(b) Y_{n-i}(a), \\ \mathcal{Q}_n^C(a, b) &= \sum_{i=1}^{n-1} [X_i(a), Y_{n-i}(b)]. \end{aligned}$$

The class $[\mathcal{Q}_n] \in \text{coker } \mathcal{D}_p$ is the order- n obstruction relative to the previously chosen lifts. Section 10 packages the entire recursion into a Maurer–Cartan equation; the Artin-ring viewpoint remains the natural language for functorial lifting problems in the sense of [2].

10. DEFORMATION DGLAS

The preceding tangent and obstruction formulas are not merely analogous to deformation theory. They arise from explicit differential graded Lie algebras. We first record a finite presentation model that reproduces the complete polynomial system, and then a more intrinsic Hochschild model for framed deformations.

10.1. A two-step presentation DGLA. Fix $p = (L, R) \in \mathfrak{A}(B)$ and write $E = \text{End}(V)$. Let

$$\mathfrak{p}_{B,p}^1 = \text{Hom}(A, E)^2$$

and let the relation space be

$$\mathcal{C}_B = \text{Hom}(A^{\otimes 3}, V)^2 \oplus E^2 \oplus \text{Hom}(A^{\otimes 2}, E)^3.$$

Set

$$\mathfrak{p}_{B,p}^2 = \mathcal{C}_B, \quad \mathfrak{p}_{B,p}^n = 0 \quad (n \neq 1, 2).$$

The differential is

$$d_p|_{\mathfrak{p}^1} = \mathcal{D}_p, \quad d_p|_{\mathfrak{p}^2} = 0.$$

For $\xi = (X, Y)$ and $\eta = (X', Y')$ in degree one, define $[\xi, \eta]_p \in \mathfrak{p}^2$ to have zero Leibniz and unit components and the following three relation components:

$$(14) \quad [\xi, \eta]_p^L(a, b) = X(a)X'(b) + X'(a)X(b),$$

$$(15) \quad [\xi, \eta]_p^R(a, b) = Y(b)Y'(a) + Y'(b)Y(a),$$

$$(16) \quad [\xi, \eta]_p^C(a, b) = [X(a), Y'(b)] + [X'(a), Y(b)].$$

Every bracket involving a degree-two element is defined to be zero.

Proposition 10.1 (Presentation DGLA). *The graded vector space*

$$\mathfrak{p}_{B,p} = \mathfrak{p}_{B,p}^1 \oplus \mathfrak{p}_{B,p}^2$$

with differential d_p and bracket $[-, -]_p$ is a two-step nilpotent differential graded Lie algebra.

Proof. Since d_p vanishes on degree two, $d_p^2 = 0$. The bracket of two degree-one elements is symmetric, as required by graded skew-symmetry in odd degree. Every nested bracket vanishes because degree two is central, so the graded Jacobi identity holds. The derivation identity for d_p is also immediate: both sides vanish after a bracket lands in degree two. $\square$

Theorem 10.2 (Exact Maurer–Cartan description). *For $\xi = (X, Y) \in \mathfrak{p}_{B,p}^1$,*

$$(L + X, R + Y) \in \mathfrak{A}(B)$$

if and only if

$$d_p \xi + \frac{1}{2}[\xi, \xi]_p = 0.$$

Consequently,

$$H^1(\mathfrak{p}_{B,p}) = T_p \mathbf{R}_B, \quad H^2(\mathfrak{p}_{B,p}) = \text{coker } \mathcal{D}_p = \text{Obs}_p(B),$$

and

$$\text{ob}_{2,p}(\xi) = \left[\frac{1}{2}[\xi, \xi]_p \right].$$

Proof. The weak Leibniz and unit equations are affine-linear and contribute only $d_p \xi$. Expanding the left representation, right antirepresentation, and commuting-action equations at (L, R) gives respectively

$$\begin{aligned}(a, b) &\longmapsto d_L X(a, b) + X(a)X(b), \\ (a, b) &\longmapsto d_R Y(a, b) + Y(b)Y(a),\end{aligned}$$

and

$$(a, b) \longmapsto \chi_{L,R}(X, Y)(a, b) + [X(a), Y(b)],$$

which are exactly the corresponding components of $d_p \xi + \frac{1}{2}[\xi, \xi]_p$. The cohomology identities follow because $\mathfrak{p}^0 = \mathfrak{p}^3 = 0$. $\square$

Remark 10.3 (Presentation dependence). *The space $H^2(\mathfrak{p}_{B,p})$ is a correct recipient for obstruction classes, but it need not be minimal. The chosen list of defining equations can contain linear dependencies and higher syzygies, which enlarge the cokernel without producing additional geometric obstructions. This motivates the intrinsic model below.*

10.2. A constrained Hochschild DGLA. Let

$$S = A^e = A \otimes A^{\text{op}}.$$

The bimodule structure (L, R) is equivalent to the algebra homomorphism

$$\mu : S \longrightarrow E, \quad \mu(a \otimes b^{\text{op}}) = L(a)R(b).$$

Regard E as an S -bimodule through μ , and let

$$C^n(S, E_\mu) = \text{Hom}(\overline{S}^{\otimes n}, E), \quad \overline{S} = S/\mathbb{C}1,$$

be the normalized Hochschild cochains. Their differential is

$$\begin{aligned}(d_\mu f)(s_1, \dots, s_{n+1}) &= \mu(s_1)f(s_2, \dots, s_{n+1}) \\ &\quad + \sum_{i=1}^n (-1)^i f(s_1, \dots, s_i s_{i+1}, \dots, s_{n+1}) \\ &\quad + (-1)^{n+1} f(s_1, \dots, s_n) \mu(s_{n+1}).\end{aligned}$$

The cup product is

$$(f \smile g)(s_1, \dots, s_{m+n}) = f(s_1, \dots, s_m)g(s_{m+1}, \dots, s_{m+n}),$$

and its graded commutator makes the Hochschild differential graded algebra into a DGLA.

For $f \in C^1(S, E_\mu)$, put

$$X_f(a) = f(a \otimes 1), \quad Y_f(b) = f(1 \otimes b^{\text{op}}),$$

and define the linear B -compatibility operator

$$\lambda_B : C^1(S, E_\mu) \longrightarrow \text{Hom}(A^{\otimes 3}, V)^2$$

by

$$\begin{aligned}\lambda_B(f)_1(a, b, c) &= X_f(a)B(b, c) + Y_f(b)B(a, c), \\ \lambda_B(f)_2(a, b, c) &= Y_f(c)B(a, b) + X_f(b)B(a, c).\end{aligned}$$

Define

$$\mathfrak{g}_{B,p}^1 = \ker \lambda_B, \quad \mathfrak{g}_{B,p}^n = C^n(S, E_\mu) \quad (n \geq 2), \quad \mathfrak{g}_{B,p}^n = 0 \quad (n \leq 0).$$

Proposition 10.4 (Constrained Hochschild DGLA). *The graded subspace $\mathfrak{g}_{B,p}$ is a DGLA. For every local Artin $\mathbb{C}$ -algebra $(R_{\text{Art}}, \mathfrak{m})$, its Maurer–Cartan elements in $\mathfrak{g}_{B,p} \otimes \mathfrak{m}$ are in bijection with framed strict reconstruction deformations of (L, R) over R_{Art} that keep A , V , and B fixed.*

Proof. The only additional condition occurs in degree one. Its differential and the bracket of two degree-one elements land in the unrestricted degree-two space, so $\mathfrak{g}_{B,p}$ is closed under both operations. For $\alpha \in C^1(S, E_\mu) \otimes \mathfrak{m}$, the map $\mu + \alpha$ is an algebra homomorphism precisely when

$$d_\mu \alpha + \alpha \smile \alpha = d_\mu \alpha + \frac{1}{2}[\alpha, \alpha] = 0.$$

Because the Leibniz equations are linear in the actions, preservation of B is exactly the condition $\lambda_B(\alpha) = 0$. $\square$

Theorem 10.5 (Intrinsic tangent and primary obstruction). *There is a natural isomorphism*

$$H^1(\mathfrak{g}_{B,p}) \cong T_p \mathbf{R}_B.$$

If $\xi = (X, Y)$ corresponds to the cocycle f_ξ , then its intrinsic primary obstruction is

$$\text{Ob}_{2,p}^{\text{Hoch}}(\xi) = [f_\xi \smile f_\xi] \in H^2(\mathfrak{g}_{B,p}),$$

and ξ extends to second order if and only if this class vanishes.

Proof. If $d_\mu f = 0$, then

$$f(a \otimes b^{\text{op}}) = X_f(a)R(b) + L(a)Y_f(b),$$

so a normalized Hochschild cocycle is determined by its two restrictions. The cocycle equation on the two factor subalgebras and on mixed products yields precisely the linearized left representation, right antirepresentation, and commuting-action equations; $\lambda_B(f) = 0$ gives the two linearized Leibniz equations. Conversely, a strict tangent pair defines a cocycle by the displayed formula. Since degree zero has been removed in the framed problem, cocycles are not quotiented by infinitesimal conjugations.

For a first-order cocycle f_ξ , the coefficient of t^2 in the Maurer–Cartan equation is

$$d_\mu f_2 = -f_\xi \smile f_\xi.$$

The required correction must lie in $\mathfrak{g}_{B,p}^1$, so solvability is equivalent to vanishing of the stated class in $H^2(\mathfrak{g}_{B,p})$. $\square$

10.3. A relation-level DGLA morphism. For $f \in \mathfrak{g}_{B,p}^1$, define

$$\Theta(f) = (X_f, Y_f) \in \mathfrak{p}_{B,p}^1.$$

For a Hochschild two-cochain ϕ , define

$$\Pi(\phi) = \left(\begin{array}{l} (a, b) \mapsto \phi(a \otimes 1, b \otimes 1) \\ (a, b) \mapsto \phi(1 \otimes b^{\text{op}}, 1 \otimes a^{\text{op}}) \\ (a, b) \mapsto \phi(a \otimes 1, 1 \otimes b^{\text{op}}) - \phi(1 \otimes b^{\text{op}}, a \otimes 1) \end{array} \right).$$

Let

$$\iota_{\text{rel}} : \text{Hom}(A^{\otimes 2}, E)^3 \longrightarrow \mathcal{C}_B$$

insert zero weak-Leibniz and unit components. Define graded maps

$$\Psi^1 = \Theta, \quad \Psi^2 = \iota_{\text{rel}} \circ \Pi, \quad \Psi^n = 0 \quad (n \geq 3).$$

Proposition 10.6 (Relation DGLA morphism). *The maps Ψ^n define a morphism of DGLAs*

$$\Psi : \mathfrak{g}_{B,p} \longrightarrow \mathfrak{p}_{B,p}.$$

Proof. For $f \in \mathfrak{g}_{B,p}^1$, normalizedness gives $X_f(1) = Y_f(1) = 0$, while $\lambda_B(f) = 0$ gives the two linearized weak-Leibniz components. On the two factor subalgebras and on mixed products, direct expansion gives

$$\begin{aligned}\Pi(d_\mu f)_L(a, b) &= d_L X_f(a, b), \\ \Pi(d_\mu f)_R(a, b) &= d_R Y_f(a, b), \\ \Pi(d_\mu f)_C(a, b) &= \chi_{L,R}(X_f, Y_f)(a, b).\end{aligned}$$

Hence

$$\Psi^2(d_\mu f) = d_p \Psi^1(f).$$

The target differential vanishes on degree two and Ψ vanishes in degrees at least three, so this proves compatibility with the differential in all degrees.

For $f, g \in \mathfrak{g}_{B,p}^1$, the degree-one Hochschild bracket is

$$[f, g] = f \smile g + g \smile f.$$

Applying Π gives

$$\begin{aligned}\Pi([f, g])_L(a, b) &= X_f(a)X_g(b) + X_g(a)X_f(b), \\ \Pi([f, g])_R(a, b) &= Y_f(b)Y_g(a) + Y_g(b)Y_f(a), \\ \Pi([f, g])_C(a, b) &= [X_f(a), Y_g(b)] + [X_g(a), Y_f(b)],\end{aligned}$$

which is precisely $[\Theta(f), \Theta(g)]_p$. Every bracket involving degree two maps to zero on both sides. Thus Ψ is a DGLA morphism. $\square$

Corollary 10.7 (Cohomological obstruction projection). *The DGLA morphism Ψ induces natural maps*

$$H^1(\Psi) : H^1(\mathfrak{g}_{B,p}) \longrightarrow H^1(\mathfrak{p}_{B,p}), \quad \bar{\Pi} := H^2(\Psi) : H^2(\mathfrak{g}_{B,p}) \longrightarrow \text{coker } \mathcal{D}_p.$$

Under the tangent-space identifications, $H^1(\Psi)$ is the identity on $T_p \mathbf{R}_B$. Moreover, for every $\xi \in T_p \mathbf{R}_B$,

$$\boxed{\bar{\Pi}(\text{Ob}_{2,p}^{\text{Hoch}}(\xi)) = \text{ob}_{2,p}(\xi).}$$

Equivalently, the diagram

$$\begin{array}{ccc} T_p \mathbf{R}_B & \xrightarrow{\text{Ob}_{2,p}^{\text{Hoch}}} & H^2(\mathfrak{g}_{B,p}) \\ \text{id} \downarrow & & \downarrow \bar{\Pi} \\ T_p \mathbf{R}_B & \xrightarrow{\text{ob}_{2,p}} & \text{coker } \mathcal{D}_p \end{array}$$

commutes.

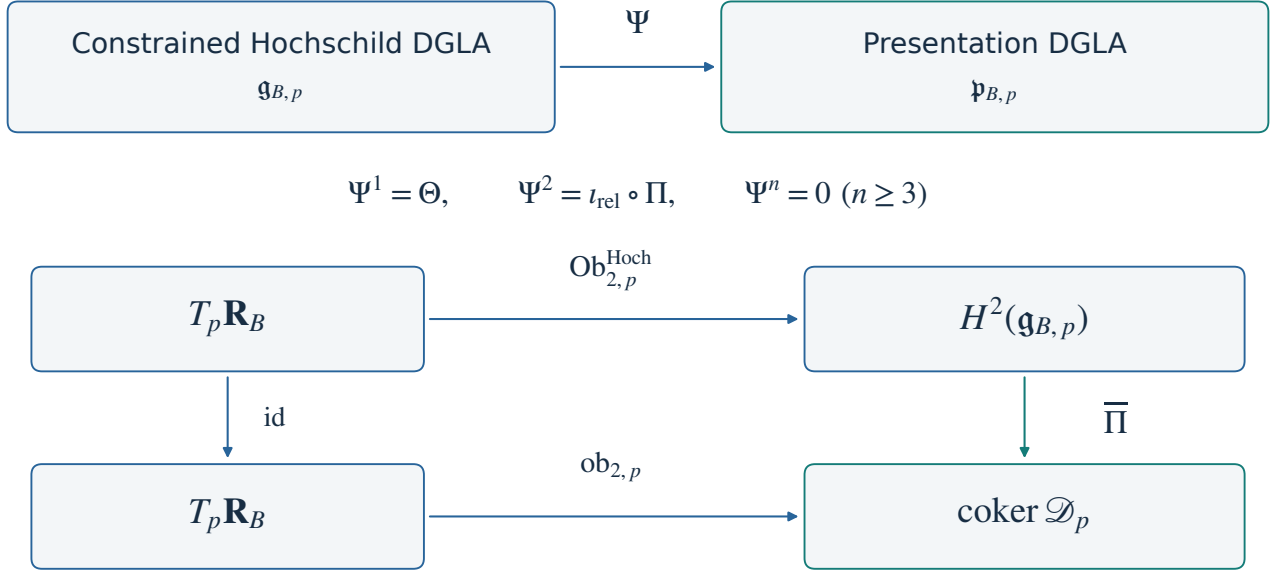
Proof. The first assertion follows from Theorems 10.2 and 10.5 together with the degree-one definition of Ψ . If f_ξ represents $\xi = (X, Y)$, then

$$\Pi(f_\xi \smile f_\xi) = \begin{pmatrix} (a, b) \mapsto X(a)X(b) \\ (a, b) \mapsto Y(b)Y(a) \\ (a, b) \mapsto [X(a), Y(b)] \end{pmatrix},$$

which is exactly the nonzero relation part of $\mathcal{D}_p(\xi)$. Passing to cohomology gives the claimed identity. $\square$

The corresponding structures are illustrated in Figure 5.

The relation morphism preserves the differential, bracket and quadratic obstruction.



The commuting square compares obstruction classes; the two DGLAs need not be quasi-isomorphic.

FIGURE 5. The relation DGLA morphism and its induced commuting obstruction square (Proposition 10.6 and Corollary 10.7). The map on tangent spaces is the identity; the degree-two map projects intrinsic obstruction classes to presentation relations. No quasi-isomorphism is asserted.

10.4. Exact cohomology and the minimal Kuranishi sector for the dual-number example. For the example of Section 8, write

$$x = \epsilon \otimes 1, \quad y = 1 \otimes \epsilon^{\text{op}}, \quad w = xy.$$

Then $\bar{S}$ has basis x, y, w , with the only nonzero products in $\bar{S}$ given by

$$xy = yx = w.$$

Let δ_{uv} denote the normalized two-cochain that takes value 1 on (u, v) and vanishes on the other basis pairs.

The B -compatibility condition on a normalized one-cochain is

$$f(x) + f(y) = 0.$$

Thus $\mathfrak{g}_{B,p}^1$ has basis

$$\tau = (-1, 1, 0), \quad \nu = (0, 0, 1),$$

where the triples record the values on (x, y, w) . The Hochschild differential satisfies

$$d_\mu \tau = 0, \quad d_\mu \nu = \beta := -(\delta_{xy} + \delta_{yx}).$$

Consequently,

$$H^1(\mathfrak{g}_{B,p}) = \mathbb{C}[\tau].$$

Exact calculation gives

$$Z^2(\mathfrak{g}_{B,p}) = \text{span}\{\eta, \eta_1, \eta_2, \beta\},$$

where

$$\begin{aligned}\eta &= \tau \smile \tau = \delta_{xx} - \delta_{xy} - \delta_{yx} + \delta_{yy}, \\ \eta_1 &= \delta_{xx} - \delta_{yy}, \\ \eta_2 &= \delta_{xy} - \delta_{yx}.\end{aligned}$$

Since $B^2 = \mathbb{C}\beta$, one obtains

$$H^2(\mathfrak{g}_{B,p}) = \mathbb{C}[\eta] \oplus \mathbb{C}[\eta_1] \oplus \mathbb{C}[\eta_2].$$

In particular,

$$\dim \mathfrak{g}_{B,p}^1 = 2, \quad \dim Z^1 = 1, \quad \dim Z^2 = 4, \quad \dim H^2 = 3.$$

Proposition 10.8 (The intrinsic relation map is injective in degree two). *For the dual-number example, the map*

$$\overline{\Pi} : H^2(\mathfrak{g}_{B,p}) \longrightarrow \text{coker } \mathcal{D}_p$$

is injective.

Proof. Restrict to the three left, right, and mixed relation coordinates evaluated at (ϵ, ϵ) . The three cohomology generators have images

$$\overline{\Pi}[\eta] = (1, 1, 0), \quad \overline{\Pi}[\eta_1] = (1, -1, 0), \quad \overline{\Pi}[\eta_2] = (0, 0, 2).$$

At the origin $L(\epsilon) = R(\epsilon) = 0$, and $\epsilon^2 = 0$, so the linearized left and right representation equations vanish identically in these two (ϵ, ϵ) coordinates. The mixed coordinate also vanishes on $\text{im } \mathcal{D}_p$ because $E = \text{End}(\mathbb{C}) = \mathbb{C}$ is commutative. The three displayed vectors therefore remain linearly independent in the cokernel. $\square$

10.5. Contraction and transfer conventions. A contraction of a cochain complex onto its cohomology means chain maps i, π and a degree -1 map h with

$$\pi i = I, \quad dh + hd = I - i\pi, \quad h^2 = hi = \pi h = 0.$$

All strict-transfer statements use these standard normalized linear splittings. For a degree-one cohomology parameter x , write

$$F_1 = ix, \quad R_n = \frac{1}{2} \sum_{\substack{a+b=n \\ a,b \geq 1}} [F_a, F_b], \quad F_n = -hR_n, \quad \kappa_n = \pi R_n \quad (n \geq 2).$$

This is the degree-one rooted-tree transfer recursion [11, Chapter 10]; the usual Taylor coefficients differ by factorial normalization. In particular, π acts on all degree-two cochains, not only on cocycles. A formal change of Kuranishi coordinates or relation generators is not itself a change of these linear splitting data.

Theorem 10.9 (Minimal Kuranishi equation for the false tangent). *There is a homotopy transfer of $\mathfrak{g}_{B,p}$ to its cohomology for which the operations on the H^1 -generated Maurer–Cartan sector satisfy*

$$\ell_2([\tau], [\tau]) = 2[\eta], \quad \ell_n([\tau], \dots, [\tau]) = 0 \quad (n \geq 3).$$

Hence the minimal Kuranishi map is

$$\kappa(t[\tau]) = t^2[\eta].$$

For every local Artin algebra $(R_{\text{Art}}, \mathfrak{m})$, the framed deformation functor of the dual-number reconstruction point is therefore

$$\text{MC}(R_{\text{Art}}) \cong \{t \in \mathfrak{m} \mid t^2 = 0\},$$

and its prorepresenting hull is

$$\boxed{\mathbb{C}[[t]]/(t^2)}.$$

This agrees with the completed local ring of

$$\mathbf{R}_B \cong \operatorname{Spec} \mathbb{C}[\ell]/(\ell^2)$$

at its unique geometric point.

Proof. We use the rooted-tree form of homotopy transfer [11, Chapter 10]. Choose representatives of H^2 to be η, η_1, η_2 , and choose a degree -1 homotopy with

$$h(\beta) = \nu, \quad h(\eta) = h(\eta_1) = h(\eta_2) = 0.$$

Since all spaces are vector spaces over $\mathbb{C}$, this low-degree splitting extends degree-wise to a contraction of the full cochain complex onto its cohomology. The transferred binary bracket is the projection of the original DGLA bracket, so

$$\ell_2([\tau], [\tau]) = \pi[\tau, \tau] = 2[\tau \smile \tau] = 2[\eta].$$

Moreover,

$$h[\tau, \tau] = 2h(\eta) = 0.$$

Every rooted binary tree contributing to $\ell_n([\tau], \dots, [\tau])$ for $n \geq 3$ contains a lowest internal edge carrying the factor $h[\tau, \tau]$; hence every such tree vanishes. The minimal Maurer–Cartan equation is therefore

$$\frac{1}{2}\ell_2(t[\tau], t[\tau]) = t^2[\eta] = 0.$$

Since $[\eta] \neq 0$, this is equivalent to $t^2 = 0$. The final identification follows from the explicit strict reconstruction equation of Section 8. $\square$

Remark 10.10 (Cohomological receptacle versus effective obstruction image). *The equality $\dim H^2 = 3$ does not mean that the one-dimensional local germ requires three equations. Proposition 10.8 shows that all three classes are genuine independent relation-level classes, not projection artifacts. Nevertheless, the Kuranishi map from the one-dimensional tangent space has image*

$$\operatorname{im} \kappa = \mathbb{C}[\eta].$$

Thus one must distinguish the full obstruction receptacle H^2 from the effective obstruction directions actually reached by the Maurer–Cartan equation near a specified point.

Remark 10.11 (When higher L_∞ operations become necessary). *For the dual-number point, Theorem 10.9 shows that all higher operations vanish on the H^1 -generated Maurer–Cartan sector. In larger reconstruction problems, nonzero higher brackets may appear after homotopy transfer, taking a homotopy fiber or gauge quotient, or allowing the multiplication of A and the map B to deform simultaneously.*

11. EFFECTIVE OBSTRUCTION SPACES AND MINIMAL EQUATIONS

The preceding examples show that H^2 is generally too large to count the equations of the local reconstruction germ. This section isolates the intrinsic finite-dimensional quotient that does count them.

Assume that $T = H^1(\mathfrak{g}_{B,p})$ and $O = H^2(\mathfrak{g}_{B,p})$ are finite-dimensional, and choose a transferred minimal L_∞ structure on cohomology. Its Maurer–Cartan equation on degree one is the formal Kuranishi map

$$\kappa(\xi) = \sum_{n \geq 2} \frac{1}{n!} \ell_n(\xi, \dots, \xi) \in O \widehat{\otimes} \widehat{\text{Sym}}(T^\vee).$$

Put

$$P_T = \widehat{\text{Sym}}(T^\vee), \quad \mathfrak{n} = P_T^{\geq 1},$$

and let

$$\kappa^\sharp : O^\vee \longrightarrow \mathfrak{n}^2, \quad \lambda \longmapsto \lambda \circ \kappa.$$

The *Kuranishi ideal* is

$$I_\kappa = P_T \kappa^\sharp(O^\vee).$$

The framed formal reconstruction germ is represented by

$$R_p = P_T / I_\kappa.$$

Because the transferred structure is minimal, $I_\kappa \subseteq \mathfrak{n}^2$, so this is a minimal formal embedding: the cotangent space of R_p is $T^\vee$.

Definition 11.1 (Effective relation and obstruction spaces). *The effective relation space and effective obstruction space at p are*

$$\mathcal{R}_p^{\text{eff}} := I_\kappa / \mathfrak{n}I_\kappa, \quad \text{Obs}_p^{\text{eff}} := (\mathcal{R}_p^{\text{eff}})^\vee.$$

Proposition 11.2 (Minimal-equation theorem). *The number*

$$\dim_{\mathbb{C}} \text{Obs}_p^{\text{eff}} = \dim_{\mathbb{C}} I_\kappa / \mathfrak{n}I_\kappa$$

is the minimum number of formal equations required to generate the defining ideal of the framed reconstruction germ inside a smooth formal space of minimal embedding dimension $\dim T$.

Proof. A family $f_1, \dots, f_r \in I_\kappa$ generates I_κ if and only if its residue classes span $I_\kappa / \mathfrak{n}I_\kappa$. This is the complete-local form of Nakayama’s lemma. Thus every basis of $I_\kappa / \mathfrak{n}I_\kappa$ lifts to a minimal generating family, and no smaller family can generate the ideal. $\square$

Theorem 11.3 (Independence of the minimal Kuranishi presentation). *Let R be a complete Noetherian local $\mathbb{C}$ -algebra with residue field $\mathbb{C}$. Suppose*

$$P = \mathbb{C}[[x_1, \dots, x_e]] \twoheadrightarrow R, \quad P' = \mathbb{C}[[y_1, \dots, y_e]] \twoheadrightarrow R$$

are two minimal presentations, with kernels $I \subseteq \mathfrak{n}^2$ and $I' \subseteq (\mathfrak{n}')^2$. Then a formal change of coordinates identifies the presentations and induces an isomorphism

$$I / \mathfrak{n}I \cong I' / \mathfrak{n}'I'.$$

Consequently, the isomorphism class of $\mathcal{R}_p^{\text{eff}}$, the isomorphism class of $\text{Obs}_p^{\text{eff}}$, and their common dimension are independent of the chosen contraction and minimal Kuranishi presentation.

Proof. Both sets of variables map to minimal generating systems of the maximal ideal of R , hence their images give bases of $\mathfrak{m}_R / \mathfrak{m}_R^2$. Lift the images of the x_i to formal series in the y_j . This defines a continuous homomorphism

$$\varphi : P \longrightarrow P'$$

commuting with the maps to R . Its linear part is the change-of-basis matrix between two bases of $\mathfrak{m}_R/\mathfrak{m}_R^2$, hence is invertible. The formal inverse function theorem makes φ an isomorphism. Since the two maps to R commute with φ , one has $\varphi(I) = I'$, and therefore

$$I/\mathfrak{n}I \xrightarrow{\sim} I'/\mathfrak{n}'I'.$$

Applying this to two minimal Kuranishi presentations proves the final assertion. $\square$

Remark 11.4. *The preceding proof uses only minimal formal power-series presentations and the formal inverse function theorem. No completed or continuous cotangent-complex identification is needed for the statements below.*

Proposition 11.5 (Embedding into the cohomological obstruction receptacle). *The component map of the Kuranishi equation induces a natural surjection*

$$q_\kappa : O^\vee \twoheadrightarrow \mathcal{R}_p^{\text{eff}}, \quad \lambda \longmapsto [\lambda \circ \kappa].$$

Hence, when O is finite-dimensional, duality gives an injection

$$j_\kappa : \text{Obs}_p^{\text{eff}} \hookrightarrow O = H^2(\mathfrak{g}_{B,p}).$$

The abstract effective obstruction space is intrinsic by Theorem 11.3; its displayed embedding into a fixed coordinate realization of H^2 is determined only up to the linear isomorphism induced by a change of minimal model.

Proof. The ideal I_κ is generated by the image of $\kappa^\sharp$, so the residue classes of those components span $I_\kappa/\mathfrak{n}I_\kappa$. This proves surjectivity. Dualizing a surjection of finite-dimensional vector spaces gives the injection j_κ . $\square$

Combining Proposition 11.5 with Corollary 10.7 gives the three-stage comparison

$$\boxed{\text{Obs}_p^{\text{eff}} \xrightarrow{j_\kappa} H^2(\mathfrak{g}_{B,p}) \xrightarrow{\overline{\Pi}} \text{coker } \mathcal{D}_p.}$$

We call

$$\text{Obs}_p^{\text{rel,eff}} := \text{im}(\overline{\Pi} \circ j_\kappa)$$

the effective relation-level obstruction space. The map

$$\overline{\Pi}_{\text{eff}} := \overline{\Pi} \circ j_\kappa$$

need not be injective in general; its kernel and the complement of its image in $\text{coker } \mathcal{D}_p$ measure different failures and should not both be called presentation syzygies.

Corollary 11.6 (Effective obstruction for the dual-number germ). *For the dual-number example,*

$$I_\kappa = (t^2), \quad \mathcal{R}_p^{\text{eff}} = (t^2)/(t^3), \quad \dim \text{Obs}_p^{\text{eff}} = 1.$$

Under the basis of Section 10, the injection j_κ has image

$$\text{im } j_\kappa = \mathbb{C}[\eta] \subset H^2(\mathfrak{g}_{B,p}).$$

Moreover, $\overline{\Pi}_{\text{eff}}$ is injective.

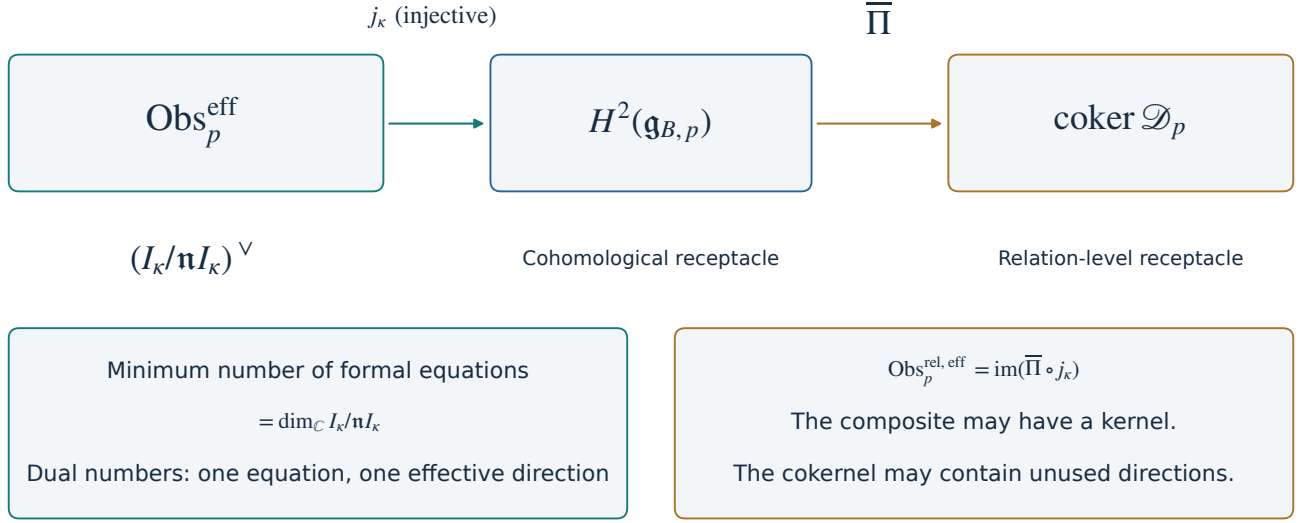
Proof. The first three assertions follow immediately from $I_\kappa = (t^2)$. Since

$$\kappa(t[\tau]) = t^2[\eta],$$

the dual component map kills $\eta_1^\vee$ and $\eta_2^\vee$ and maps $\eta^\vee$ to the generator of $(t^2)/(t^3)$. Thus the dual injection lands on $\mathbb{C}[\eta]$. Proposition 10.8 then implies injectivity after applying $\overline{\Pi}$. $\square$

The corresponding structures are illustrated in Figure 6.

Minimal equations, cohomology and presentation relations occupy different levels.



The abstract effective space is intrinsic; its displayed embedding depends on the minimal model.

FIGURE 6. The effective obstruction comparison chain. The dimension of $(I_\kappa / \mathfrak{n}I_\kappa)^\vee$ counts minimal formal equations. Its displayed injection into H^2 depends on the minimal-model realization, and its composite with $\bar{\Pi}$ need not be injective. Unused relation directions and the kernel of the composite are different phenomena.

11.1. A two-parameter mixed-obstruction example. Let

$$A = \mathbb{C}1 \oplus \mathbb{C}a \oplus \mathbb{C}b, \quad (a, b)^2 = 0,$$

and let $V = \mathbb{C}^2$ with basis e_1, e_2 . Define

$$B(b, a) = e_1, \quad B(b, b) = e_2,$$

and let all other values on basis pairs vanish. Then $V = \text{span } B(A, A)$. Let p be the augmentation bimodule structure

$$L(x) = R(x) = \varepsilon(x)I_V.$$

Because B vanishes whenever one input is scalar and the radical of A has square zero, B is a biderivation relative to p .

Proposition 11.7 (Exact strict reconstruction scheme). *Every unit-preserving weak reconstruction has*

$$L(a) = R(a) = 0, \quad L(b) = N(u, v), \quad R(b) = -N(u, v),$$

where

$$N(u, v) = \begin{pmatrix} 0 & -u \\ 0 & -v \end{pmatrix}.$$

It is strict precisely when

$$uv = 0, \quad v^2 = 0.$$

Consequently,

$$\mathbf{R}_B \cong \text{Spec } \frac{\mathbb{C}[u, v]}{(uv, v^2)}.$$

Proof. Evaluating the two weak Leibniz identities on the eight triples from $\{a, b\}^3$ gives a linear system in the sixteen matrix entries of

$$L(a), L(b), R(a), R(b).$$

Exact row reduction has rank fourteen and yields precisely the displayed two-parameter solution. Since the radical of A has square zero, all left and right representation equations reduce to $N(u, v)^2 = 0$; the mixed equations are automatic because $R(b) = -L(b)$. Finally,

$$N(u, v)^2 = \begin{pmatrix} 0 & uv \\ 0 & v^2 \end{pmatrix},$$

which gives the stated ideal. $\square$

Remark 11.8 (A smooth branch with an embedded false direction). *The reduced reconstruction scheme is the affine line $v = 0$. The u -direction integrates to an actual family of strict reconstructions, while the v -direction exists in the Zariski tangent space only at the origin and is nilpotently embedded there. The mixed equation $uv = 0$ records the failure to combine the genuine u -deformation with the false v -direction.*

For the intrinsic complex, put

$$x = b \otimes 1, \quad y = 1 \otimes b^{\text{op}},$$

and define matrices

$$P = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}.$$

Define normalized one-cochains α, β by

$$\alpha(x) = P, \quad \alpha(y) = -P, \quad \beta(x) = Q, \quad \beta(y) = -Q,$$

and let them vanish on the other normalized basis elements of A^e .

Theorem 11.9 (Two-parameter minimal Kuranishi equation). *For the square-zero example,*

$$\begin{aligned} \dim \mathfrak{g}_{B,p}^1 &= 18, & \dim H^1(\mathfrak{g}_{B,p}) &= 2, \\ \dim Z^2(\mathfrak{g}_{B,p}) &= 64, & \dim H^2(\mathfrak{g}_{B,p}) &= 48. \end{aligned}$$

The classes $[\alpha], [\beta]$ form a basis of H^1 . Put

$$\eta_{uv} = \alpha \smile \beta + \beta \smile \alpha, \quad \eta_{vv} = \beta \smile \beta.$$

Then $[\eta_{uv}]$ and $[\eta_{vv}]$ are nonzero and linearly independent in H^2 . There is a homotopy transfer for which

$$\begin{aligned} \ell_2([\alpha], [\alpha]) &= 0, \\ \ell_2([\alpha], [\beta]) &= [\eta_{uv}], \\ \ell_2([\beta], [\beta]) &= 2[\eta_{vv}], \end{aligned}$$

and every ℓ_n with $n \geq 3$ vanishes on the H^1 -generated Maurer–Cartan sector. Hence

$$\kappa(u[\alpha] + v[\beta]) = uv[\eta_{uv}] + v^2[\eta_{vv}].$$

The Kuranishi ideal is (uv, v^2) , in agreement with Proposition 11.7.

Proof. The normalized augmentation ideal of $S = A^e$ has dimension eight. Exact rational linear algebra gives

$$\text{rank } \lambda_B = 14, \quad \text{rank}(d_1|_{\mathfrak{g}_{B,p}^1}) = 16, \quad \text{rank } d_2 = 192,$$

which yields the four displayed dimensions. Direct substitution shows that α and β are B -compatible cocycles and $\text{span } Z^1 = H^1$.

One has

$$\alpha \smile \alpha = 0.$$

The two cochains η_{uv} and η_{vv} are closed, each increases the boundary rank by one, and together increase it by two; hence their classes are nonzero and independent. Choose them as cohomology representatives and choose the contracting homotopy to vanish on both. Every higher transfer tree on $[\alpha], [\beta]$ then has a lowest internal edge carrying

$$h[\alpha, \alpha], \quad h[\alpha, \beta], \quad \text{or} \quad h[\beta, \beta],$$

all of which vanish. Thus only the binary bracket contributes, and

$$\frac{1}{2}\ell_2(u[\alpha] + v[\beta], u[\alpha] + v[\beta]) = uv[\eta_{uv}] + v^2[\eta_{vv}].$$

The agreement with the exact reconstruction scheme completes the proof. $\square$

Corollary 11.10 (Forty-eight-dimensional receptacle, two effective equations). *For the square-zero example,*

$$\dim H^2(\mathfrak{g}_{B,p}) = 48, \quad \dim \text{Obs}_p^{\text{eff}} = 2,$$

and

$$\text{im } j_\kappa = \text{span}\{[\eta_{uv}], [\eta_{vv}]\}.$$

The effective relation map $\overline{\Pi}_{\text{eff}}$ is injective.

Proof. Since $I_\kappa = (uv, v^2)$ and $\mathfrak{n} = (u, v)$, the classes of uv and v^2 form a basis of

$$I_\kappa / \mathfrak{n}I_\kappa.$$

The Kuranishi component map identifies the dual effective space with the span of the two stated cohomology classes. Under the relation projection, their left representation components at (b, b) are respectively

$$PQ + QP = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Q^2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The linearized left relation at (b, b) vanishes identically because $b^2 = 0$ and $L(b) = 0$ at the base point. These two matrix coordinates therefore remain independent in $\text{coker } \mathcal{D}_p$. $\square$

12. A FIRST GENUINELY CUBIC KURANISHI OBSTRUCTION

The preceding nonreduced examples are quadratically controlled. We now give an explicit inverse-Leibniz reconstruction in which the complete quadratic Kuranishi map on the strict tangent space vanishes, but the cubic term does not.

Let

$$A = \mathbb{C}[\epsilon]/(\epsilon^4), \quad V = \mathbb{C}^3,$$

and let v_1, v_2, v_3 be the standard basis of V . Put

$$N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Define a strict base reconstruction $p = (L, R)$ by

$$L(\epsilon) = N, \quad R(\epsilon) = N + N^2,$$

with the values on higher powers determined multiplicatively. Thus

$$L(\epsilon^2) = R(\epsilon^2) = N^2, \quad L(\epsilon^3) = R(\epsilon^3) = 0.$$

The two actions commute because they are polynomials in N .

Define $B : A \times A \rightarrow V$ by declaring all values with a unit input to vanish and taking the following as its only nonzero values on radical basis pairs:

$$\begin{aligned} B(\epsilon, \epsilon) &= 3v_1 + v_2 + v_3, \\ B(\epsilon, \epsilon^2) &= B(\epsilon^2, \epsilon) = 3v_1 + 2v_2, \\ B(\epsilon, \epsilon^3) &= B(\epsilon^3, \epsilon) = 3v_1, \\ B(\epsilon^2, \epsilon^2) &= 4v_1. \end{aligned}$$

Direct substitution verifies both Leibniz identities relative to p . Moreover,

$$V = \text{span } B(A, A).$$

Let $\mathcal{K}_B$ denote the vector space of unit-preserving weak perturbations at p , and let

$$\mathcal{D}_p|_{\mathcal{K}_B} : \mathcal{K}_B \longrightarrow \mathcal{C}_B$$

be the strict tangent operator restricted to that weak space.

Proposition 12.1 (Exact weak and strict tangent dimensions). *For the preceding reconstruction,*

$$\dim \mathcal{K}_B = 9, \quad \text{rank}(\mathcal{D}_p|_{\mathcal{K}_B}) = 7, \quad \dim T_p \mathbf{R}_B = 2.$$

A basis of the strict tangent space is given by $\alpha = (X_\alpha, Y_\alpha)$ and $\xi = (X_\xi, Y_\xi)$, where

$$X_\alpha(\epsilon) = -N^2, \quad Y_\alpha(\epsilon) = N^2,$$

and all their values on ϵ^2, ϵ^3 vanish, while

$$\begin{aligned} X_\xi(\epsilon) &= -N, & X_\xi(\epsilon^2) &= -2N^2, & X_\xi(\epsilon^3) &= 0, \\ Y_\xi(\epsilon) &= N, & Y_\xi(\epsilon^2) &= 2N^2, & Y_\xi(\epsilon^3) &= 0. \end{aligned}$$

Proof. The two linearized Leibniz blocks form an exact rational matrix with fifty-four columns. Its rank is forty-five, giving the nine-dimensional weak perturbation space. Restricting the strict relation differential to this kernel gives a matrix of rank seven. The two displayed perturbations lie in both kernels and are linearly independent, hence form a tangent basis. $\square$

Write

$$\mathcal{P}_p(\rho, \sigma) := \mathcal{Q}_p(\rho + \sigma) - \mathcal{Q}_p(\rho) - \mathcal{Q}_p(\sigma)$$

for the symmetric polarization of the quadratic relation map. Define a weak second-order correction $\eta = (U, W)$ by

$$\begin{aligned} U(\epsilon) &= I, & U(\epsilon^2) &= 2N + N^2, & U(\epsilon^3) &= 3N^2, \\ W(\epsilon) &= -I, & W(\epsilon^2) &= -2N - N^2, & W(\epsilon^3) &= -3N^2. \end{aligned}$$

Then exact substitution gives

$$(17) \quad \mathcal{Q}_p(\alpha) = 0, \quad \mathcal{P}_p(\alpha, \xi) = 0, \quad \mathcal{D}_p \eta = -\mathcal{Q}_p(\xi).$$

Thus every quadratic Kuranishi product on $T_p \mathbf{R}_B$ vanishes in the obstruction cokernel.

At third order define

$$\zeta := \mathcal{P}_p(\xi, \eta) \in \mathcal{C}_B.$$

For a relation vector Θ , let $\Theta^L(a, b)$ and $\Theta^R(a, b)$ denote its left and right representation components. Consider the linear functional

$$(18) \quad \Lambda(\Theta) = 4\Theta^L(\epsilon, \epsilon)_{12} - 2\Theta^L(\epsilon, \epsilon)_{23} - 2\Theta^L(\epsilon, \epsilon^2)_{13} - 2\Theta^R(\epsilon, \epsilon)_{23}.$$

Theorem 12.2 (Nonzero cubic Kuranishi term). *For the reconstruction above,*

$$\Lambda \circ \mathcal{D}_p|_{\mathcal{K}_B} = 0, \quad \Lambda(\zeta) = 8.$$

Consequently,

$$[\zeta] \neq 0 \quad \text{in} \quad \text{coker}(\mathcal{D}_p|_{\mathcal{K}_B}).$$

For a transferred minimal model of the presentation DGLA, the homogeneous Kuranishi terms on

$$H^1 = \mathbb{C}[\alpha] \oplus \mathbb{C}[\xi]$$

may therefore be normalized as

$$\boxed{\kappa_2 = 0, \quad \kappa_3(u[\alpha] + v[\xi]) = v^3[\zeta].}$$

In particular,

$$\kappa_2([\xi], [\xi]) = 0, \quad \kappa_3([\xi], [\xi], [\xi]) \neq 0.$$

Proof. Equation (17) proves that the binary bracket of every pair of tangent classes is zero in cohomology. For the cubic term, direct matrix substitution into the relation differential gives the two identities involving Λ ; hence ζ cannot be a strict tangent boundary.

It remains to check the mixed cubic coefficient. Define $\theta = (S, T)$ by

$$\begin{aligned} S(\epsilon) &= -2I + N, & S(\epsilon^2) &= -4N, & S(\epsilon^3) &= -6N^2, \\ T(\epsilon) &= 2I - N, & T(\epsilon^2) &= 4N, & T(\epsilon^3) &= 6N^2. \end{aligned}$$

Then $\theta \in \mathcal{K}_B$ and

$$\mathcal{D}_p\theta = -\mathcal{P}_p(\alpha, \eta).$$

For the tangent vector $u\alpha + v\xi$, the second-order correction may therefore be chosen as $v^2\eta$, and its cubic source is

$$uv^2\mathcal{P}_p(\alpha, \eta) + v^3\mathcal{P}_p(\xi, \eta).$$

The first summand is a boundary and the second represents $v^3[\zeta]$. This is precisely the homogeneous cubic part of the minimal Kuranishi map.

Finally, the nonzero class does not depend on the chosen second-order lift. Any other lift differs from η by $s\alpha + t\xi$, and

$$\mathcal{P}_p(\xi, \alpha) = 0, \quad \mathcal{P}_p(\xi, \xi) = 2\mathcal{D}_p(\xi) \in \text{im } \mathcal{D}_p.$$

Thus the resulting cubic class is unchanged. $\square$

Corollary 12.3 (Second-order extension with third-order failure). *The tangent direction ξ extends to a strict reconstruction over*

$$\mathbb{C}[t]/(t^3)$$

by

$$p_t = p + t\xi + t^2\eta,$$

but it admits no extension to $\mathbb{C}[t]/(t^4)$.

Proof. The order-two equation is (17). Any order-three correction would require ζ to lie in the image of $\mathcal{D}_p|_{\mathcal{K}_B}$, contradicting Theorem 12.2. $\square$

12.1. Complete formal germ and the higher Kuranishi tail. The cubic obstruction can in fact be completed exactly. Choose the exact nine weak-fiber coordinates

$$w_0, \dots, w_8$$

coming from a nullspace basis of the weak observability matrix, normalized so that

$$\alpha = (1, 0, 0, 0, 0, 0, 0, 0, 0),$$

$$\xi = (0, 1, 0, 2, 0, 0, 0, 0, 0).$$

in these coordinates. Let u, v be tangent coordinates. Let a, b, c, d, e, f, g denote transverse coordinates, with

$$w_0 = u, \quad w_1 = v + a, \quad w_2 = b, \quad w_3 = 2v,$$

$$w_4 = c, \quad w_5 = d, \quad w_6 = e, \quad w_7 = f, \quad w_8 = g.$$

Thus the linear u - and v -axes are precisely the tangent classes $[\alpha]$ and $[\xi]$.

Theorem 12.4 (Exact local equations of the cubic example). *After restricting the full strict reconstruction equations to the weak fiber and applying the preceding linear coordinate change, an exact lexicographic Groebner basis is*

$$\boxed{\begin{array}{cccc} a + \frac{e}{6}, & b - \frac{e}{3}, & c - \frac{2e}{3}, & d, f, g, \\ e^2, & ev, & (1 + 2u)e + 3v^2, & v^3. \end{array}}$$

Consequently the local and completed local rings at the base reconstruction satisfy

$$\mathcal{O}_{\mathbf{R}_B, p} \cong \left(\frac{\mathbb{C}[u, v]}{(v^3)} \right)_{(u, v)}, \quad \hat{\mathcal{O}}_{\mathbf{R}_B, p} \cong \frac{\mathbb{C}[[u, v]]}{(v^3)}.$$

In particular,

$$I_\kappa = (v^3), \quad \dim \text{Obs}_p^{\text{eff}} = 1.$$

Proof. The exact weak equations have a nine-dimensional nullspace. Substitute its basis parameterization into every left, right, and mixed strict relation. Exact rational Groebner reduction gives the displayed basis; this is verified independently by the accompanying script.

At the origin

$$q := 1 + 2u$$

is a unit. The relation $qe + 3v^2 = 0$ therefore eliminates the transverse variable:

$$e = -\frac{3v^2}{q}.$$

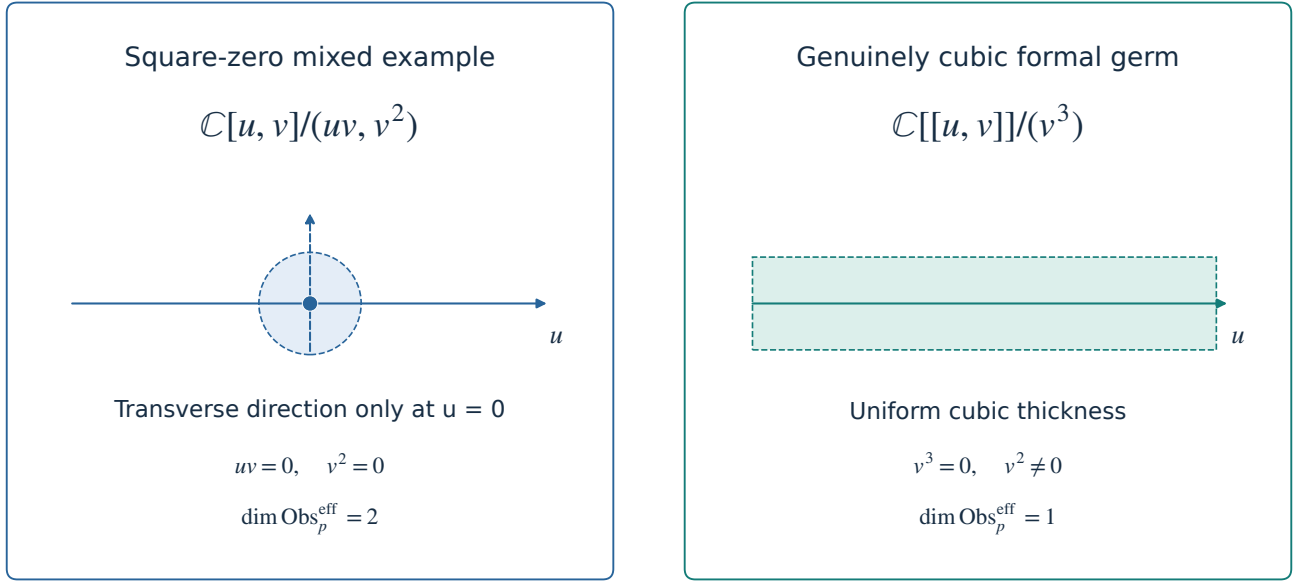
Then $ev = 0$ becomes $v^3 = 0$, while $e^2 = 0$ becomes a unit multiple of $v^4 = 0$ and is redundant modulo v^3 . The remaining transverse variables are eliminated linearly. Hence the localized ring is the localization of $\mathbb{C}[u, v]/(v^3)$ at the origin, and completion gives the stated formal ring. The final assertion follows from the minimal-equation theorem, since $(v^3)/(u, v)(v^3)$ is one-dimensional. $\square$

The corresponding structures are illustrated in Figure 7.

Proposition 12.5 (Finite flat structure of the same fixed weak fiber). *The affine strict weak-fiber scheme in Theorem 12.4 is finite flat of degree three over the u -line. More precisely, writing $q = 1 + 2u$, its coordinate algebra is*

$$\mathcal{A} = \mathbb{C}[u, e, v]/(e^2, ev, qe + 3v^2),$$

The same reduced line can carry very different transverse nilpotent structure.



Schematic support and thickness; the cubic germ is not three intersecting branches.

FIGURE 7. The mixed scheme $\text{Spec } \mathbb{C}[u, v]/(uv, v^2)$ and the cubic formal germ $\mathbb{C}[[u, v]]/(v^3)$ have different transverse nilpotent structures. The first has an embedded false direction only at the origin; the second has uniform cubic thickness. Shading is schematic and does not represent additional branches.

and is free over $\mathbb{C}[u]$ with basis $1, v, e$. On $q \neq 0$ it is

$$\mathcal{A}[q^{-1}] \simeq \mathbb{C}[u, q^{-1}, v]/(v^3).$$

At $u = -\frac{1}{2}$, its fiber is instead

$$\mathbb{C}[e, v]/(e, v)^2.$$

Thus every fiber has length three; the special fiber has embedding dimension two, whereas the other fibers have embedding dimension one.

Proof. Eliminate a, b, c, d, f, g using the displayed exact ideal. The relation $v^3 = 0$ is redundant because $3v^3 = v(qe + 3v^2) - qev$. Every monomial reduces to a $\mathbb{C}[u]$ -linear combination of $1, v, e$. To prove their independence, on a free module with that ordered basis let multiplication by v, e be represented by

$$M_v = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & -q/3 & 0 \end{pmatrix}, \quad M_e = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

They commute and satisfy $M_e^2 = M_e M_v = 0$ and $3M_v^2 + qM_e = 0$. Applying a purported relation $a + bv + ce = 0$ to the first basis vector gives $(a, b, c) = 0$. This proves freeness, and hence finite flatness. When q is invertible, eliminate $e = -3v^2/q$. When $q = 0$, the relations become $e^2 = ev = v^2 = 0$. These explicit algebras give the lengths and embedding dimensions. $\square$

The special fiber changes embedding dimension while preserving scheme-theoretic length.

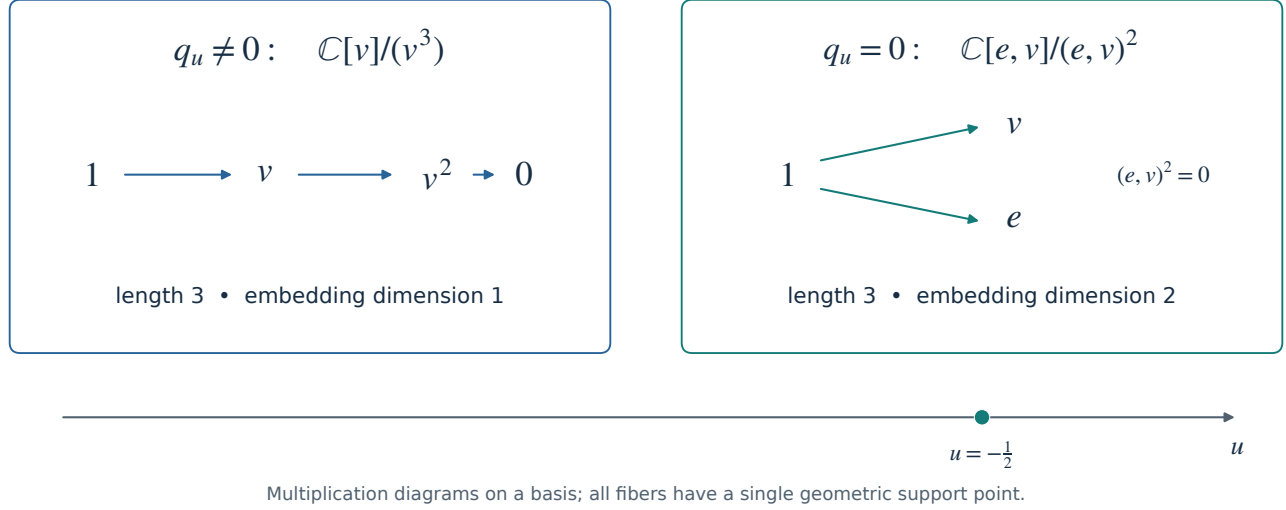


FIGURE 8. The finite flat family of Proposition 12.5. Write $q_u = 1 + 2u$, distinct from the later quintic $q(S)$. Every fiber has length three. At $u = -1/2$ its embedding dimension rises from one to two. The arrows depict multiplication on bases, not distinct geometric points.

Remark 12.6. *This is a refinement of the already computed fixed- B reconstruction scheme, not a deformation of the algebra A or of B . The original base point $u = 0$ lies in the open cubic region; the special fiber does not alter its completed local ring $\mathbb{C}[[u, v]]/(v^3)$.*

The corresponding structures are illustrated in Figure 8.

The theorem already proves that no new independent quartic or higher equation survives. One can go further and calculate an explicit all-orders Kuranishi representative tied to the cubic certificate (18).

Proposition 12.7 (Certificate-normalized higher Kuranishi terms). *Put $q = 1 + 2u$. Solve the seven transverse equations by*

$$e = -\frac{3v^2}{q}, \quad a = -\frac{e}{6}, \quad b = \frac{e}{3}, \quad c = \frac{2e}{3}, \quad d = f = g = 0.$$

Evaluating the full strict residual on this exact formal section and applying $\Lambda/8$ gives the scalar Kuranishi representative

$$k_\Lambda(u, v) = v^3 \frac{v + 4q}{4q^2}.$$

Its homogeneous pieces begin

$$\begin{aligned} k_3 &= v^3, \\ k_4 &= -2uv^3 + \frac{1}{4}v^4, \\ k_5 &= 4u^2v^3 - uv^4, \\ k_6 &= -8u^3v^3 + 3u^2v^4. \end{aligned}$$

More generally, for $r \geq 1$, the part of total degree $3 + r$ is

$$k_{3+r} = (-2)^r u^r v^3 + \frac{r(-2)^{r-1}}{4} u^{r-1} v^4.$$

Every one of these terms is a multiple of the cubic generator. Since

$$\frac{4q^2}{v + 4q}$$

is a unit with value 1 at the origin,

$$\frac{4q^2}{v + 4q} k_\Lambda(u, v) = v^3.$$

Thus an equivalent minimal Kuranishi presentation is exactly

$$\boxed{\tilde{\kappa}(u[\alpha] + v[\xi]) = v^3[\zeta], \quad \tilde{\kappa}_n = 0 \quad (n \geq 4).}$$

Proof. The exact formal section is obtained by solving the Groebner basis of Theorem 12.4. Direct substitution into all strict relations shows that every residual component is divisible by v^3 in $\mathbb{C}[[u, v]]$, and at least one quotient is a unit. Applying the sparse functional Λ gives

$$\frac{1}{8}\Lambda(\text{residual}) = \frac{v^3(v + 4q)}{4q^2}.$$

Expanding

$$\frac{1}{q} = \sum_{n \geq 0} (-2u)^n, \quad \frac{1}{q^2} = \sum_{n \geq 0} (n+1)(-2u)^n$$

gives the displayed homogeneous terms and the general formula. The final unit rescaling does not change the generated Kuranishi ideal and removes the whole higher tail at once. $\square$

Remark 12.8 (Intrinsic content versus presentation tail). *The nonzero cubic class is intrinsic, but the individual quartic and higher coefficients displayed above are not new obstruction classes. They depend on the chosen formal section and relation generator. Their simultaneous removal by a unit is exactly the distinction between the intrinsic minimal relation ideal (v^3) and a nonminimal all-orders representative of that same ideal.*

Remark 12.9 (Geometric shape of the germ). *The reduced local scheme is the smooth u -axis. Unlike the square-zero example of Section 11.1, where the false direction is pinned to the origin by the mixed equation $uv = 0$, the present germ is a uniform cubic thickening of the whole smooth branch:*

$$\text{Spf } \mathbb{C}[[u, v]]/(v^3) \cong \text{Spf } \mathbb{C}[[u]] \hat{\times} \text{Spec } \mathbb{C}[v]/(v^3).$$

The direction $[\xi]$ is therefore a nilpotent normal direction of order three along the branch, not an isolated embedded tangent direction.

Remark 12.10 (Formal normal form versus strict transfer). *The preceding proposition concerns equivalence of minimal Kuranishi presentations. Such an equivalence may multiply the obstruction generator by a formal unit depending on the tangent variables. A change of contraction of a fixed DGLA is more restrictive. The next result shows that, in the present cubic example, the formal pure-cubic normal form cannot be obtained by a strict homotopy transfer from any contraction.*

12.2. A quartic no-go theorem for strict transferred normal forms. Retain the tangent basis α, ξ and the weak perturbation space $\mathcal{K}_B$. Define the linear map

$$\mu_\alpha : \mathcal{K}_B \longrightarrow \text{coker}(\mathcal{D}_p|_{\mathcal{K}_B}), \quad k \longmapsto [\mathcal{P}_p(\alpha, k)].$$

If a third-order correction is restricted to $\mathcal{K}_B$, its variation changes the quartic uv^3 source by $\text{im } \mu_\alpha$. This restricted class is useful for comparison, but the universal contraction statement below requires a separate joint certificate.

Choose η and θ as above, so that

$$\mathcal{D}_p\eta = -\mathcal{D}_p(\xi), \quad \mathcal{D}_p\theta = -\mathcal{P}_p(\alpha, \eta).$$

Define the *weak-fiber quartic class*

$$\boxed{\nu_4(\alpha, \xi) := [\mathcal{P}_p(\xi, \theta)] \quad \text{in} \quad \frac{\text{coker}(\mathcal{D}_p|_{\mathcal{K}_B})}{\text{im } \mu_\alpha}.}$$

Lemma 12.11 (Choice independence of ν_4). *The class $\nu_4(\alpha, \xi)$ is independent of the choices of the second-order lift η and the mixed third-order lift θ satisfying the displayed equations.*

Proof. Any other second-order lift is

$$\eta' = \eta + s\alpha + t\xi.$$

Because

$$\mathcal{D}_p(\alpha) = 0, \quad \mathcal{P}_p(\alpha, \xi) = 0,$$

one has $\mathcal{P}_p(\alpha, \eta') = \mathcal{P}_p(\alpha, \eta)$ exactly. Hence any corresponding θ' differs from θ by a tangent vector $a\alpha + b\xi$. Therefore

$$\mathcal{P}_p(\xi, \theta') - \mathcal{P}_p(\xi, \theta) = a\mathcal{P}_p(\xi, \alpha) + b\mathcal{P}_p(\xi, \xi) = 2b\mathcal{D}_p(\xi),$$

which lies in $\text{im } \mathcal{D}_p$ by the defining equation for η . Thus the class is unchanged even before quotienting by $\text{im } \mu_\alpha$. $\square$

For a relation vector Θ , define a second sparse functional

$$(19) \quad \begin{aligned} \Gamma(\Theta) = & \Theta^L(\epsilon, \epsilon)_{12} - \Theta^L(\epsilon, \epsilon)_{22} + \Theta^L(\epsilon, \epsilon)_{23} - 2\Theta^L(\epsilon^2, \epsilon)_{13} \\ & - \Theta^R(\epsilon, \epsilon)_{12} - \Theta^R(\epsilon, \epsilon)_{23}. \end{aligned}$$

For a pair of relation vectors define the joint functional

$$(20) \quad \begin{aligned} \mathcal{J}(U, V) = & -U^L(\epsilon, \epsilon)_{11} - U^L(\epsilon, \epsilon)_{23} - U^R(\epsilon, \epsilon)_{23} \\ & + V^L(\epsilon, \epsilon)_{11} + V^L(\epsilon, \epsilon)_{21} + V^R(\epsilon, \epsilon)_{11}. \end{aligned}$$

Unlike the preceding weak-fiber certificate, this functional couples two quartic coefficients and allows an unrestricted third-order correction.

Theorem 12.12 (Strict transferred pure-cubic normal form is impossible). *The weak-fiber class $\nu_4(\alpha, \xi)$ is nonzero. Moreover, for every contraction of the full presentation DGLA onto cohomology, in the standard normalized convention of Section 10.5,*

$$\boxed{\ell_4|_{\text{Sym}^4 H^1} \neq 0.}$$

Hence no such contraction gives a strictly transferred pure-cubic degree-one Maurer–Cartan sector. This statement concerns the whole quartic operation, not the nonvanishing of one fixed monomial coefficient.

Proof. Write $D = \mathcal{D}_p$, $P = \mathcal{P}_p$, $Q = \mathcal{Q}_p$, and $z = P(\xi, \eta)$. Direct substitution into the fixed matrices gives

$$D\eta = -Q(\xi), \quad P(\alpha, \eta) = 2D\eta, \quad P(\alpha, \xi) = Q(\alpha) = 0.$$

The earlier functional satisfies $\Gamma D|_{\mathcal{K}_B} = 0$, $\Gamma P(\alpha, \mathcal{K}_B) = 0$ and $\Gamma P(\xi, \theta) = -16$, proving the first assertion.

For the unrestricted assertion, the six coordinates in (20) give the identities

$$\begin{aligned} \mathcal{J}(Dw, 0) &= \mathcal{J}(0, Dw) = 0 & (w \in \mathcal{K}_B), \\ (21) \quad \mathcal{J}(P(\alpha, K), P(\xi, K)) &= 0 & (K \in \mathfrak{p}_{B,p}^1), \\ \mathcal{J}(0, z) &= 0, \quad \mathcal{J}(-2z, Q(\eta)) = -6. \end{aligned}$$

These are finite linear identities: the weak equations supply a nine-dimensional basis for w , while the second line holds for all entries of $X_K(\epsilon), Y_K(\epsilon)$ without weak constraints. Entries at other powers and at the unit do not enter that line. The companion verifier checks the identities over $\mathbb{Q}$ and also verifies the tangent-parameter identity below symbolically.

Let (i, π, h) be any normalized contraction. In degree one its cohomology identifies with $\ker D = \mathbb{C}\alpha \oplus \mathbb{C}\xi$, since $\mathfrak{p}^0 = 0$. The transfer recursion therefore starts with

$$F_1 = u\alpha + v\xi, \quad F_2 = v^2 E_h, \quad E_h = \eta + s\alpha + t\xi.$$

Because E_h lies in $\text{im } h$, normalization gives $hDE_h = E_h$. Since $P(\alpha, E_h) = 2DE_h$, the next term has the form

$$F_3 = -2uv^2 E_h + v^3 K, \quad K = -hP(\xi, E_h).$$

No condition $K \in \mathcal{K}_B$ is imposed. The uv^3 and v^4 coefficients of the quartic source are respectively

$$U = P(\alpha, K) - 2P(\xi, E_h), \quad V = P(\xi, K) + Q(E_h).$$

Here $P(\xi, E_h) = z + 2tQ(\xi)$ and

$$Q(E_h) = Q(\eta) + 2sD\eta + tz + t^2Q(\xi).$$

Since $Q(\xi) = -D\eta$, equations (21) yield

$$\boxed{\mathcal{J}(U, V) = -6}$$

for every s, t, K .

Both U and V have zero unit and weak-Leibniz components. If either is a full presentation boundary Dw , those components force $w(1) = 0$ and $w \in \mathcal{K}_B$. Thus if both were boundaries the first line of (21) would give $\mathcal{J}(U, V) = 0$, a contradiction. Degree two is central and has zero outgoing differential in the presentation DGLA, so $\pi U, \pi V$ cannot both vanish. The quartic transferred operation is therefore nonzero. $\square$

Remark 12.13 (Why two coefficients are necessary). *The restriction to the weak fiber cannot be silently imposed on K . For example, set*

$$X_K(\epsilon) = 4(E_{21} + E_{32}), \quad X_K(\epsilon^2) = 8(E_{11} + E_{33}), \quad X_K(\epsilon^3) = 0,$$

with $Y_K = -X_K$ and zero unit values. Then $K \notin \mathcal{K}_B$ but $P(\alpha, K) = 2P(\xi, \eta)$. With $E_h = \eta$, this cancels U . These choices are compatible with a contraction: η, K are independent modulo $\ker D$, and the nonzero cubic class may be represented by $z + DK$, so that $h(z) = -K$ and $h(z + DK) = 0$. The joint certificate still gives $\mathcal{J}(0, V) = -6$.

Proposition 12.14 (Restricted weak-fiber rank check). *The image of μ_α has dimension three. Moreover,*

$$\text{rank}(\mathcal{D}_p, \mu_\alpha, \mathcal{P}_p(\xi, \theta)) = \text{rank}(\mathcal{D}_p, \mu_\alpha) + 1.$$

More explicitly, form the 243×18 coefficient matrix from the nine columns of $\mathcal{D}_p$ and the nine columns of $k \mapsto \mathcal{P}_p(\alpha, k)$, both evaluated on the same weak-fiber basis. Its rank is 10; adjoining $\mathcal{P}_p(\xi, \theta)$ gives rank 11.

Proof. These are exact rational rank computations in the nine-dimensional weak fiber. The sparse functional Γ is an independent certificate for the rank jump: it annihilates the first two summands and is nonzero on the last vector. $\square$

Remark 12.15 (Why the formal normal form still exists). *There is no contradiction with Proposition 12.7. The formal rescaling*

$$\frac{4(1 + 2u)^2}{v + 4(1 + 2u)}$$

acts on the relation generator by a tangent-dependent unit. Such a nonlinear change is permitted when passing between equivalent minimal Kuranishi presentations, but it is not generated by changing only the linear contraction data of the fixed presentation DGLA. The joint certificate $\mathcal{J}$ proves this gap for arbitrary normalized contractions; ν_4 separately records the restricted weak-fiber calculation.

Remark 12.16 (What remains open after the presentation calculation). *The strict transferred normal-form question is settled negatively for the presentation DGLA of this cubic example. It does not yet follow that the same quartic class is intrinsic to the full constrained Hochschild DGLA, because the latter has a nontrivial differential out of degree two. The next subsection resolves this distinction explicitly.*

12.3. The presentation quartic class does not lift intrinsically. Write

$$\mathbf{a}, \mathbf{x} \in \mathfrak{g}_{B,p}^1$$

for the normalized Hochschild one-cocycles corresponding to the tangent vectors α, ξ . Thus, for $a, b \in A$,

$$\mathbf{a}(a \otimes b^{\text{op}}) = X_\alpha(a)R(b) + L(a)Y_\alpha(b),$$

and similarly for $\mathbf{x}$. Let $\eta = (U, W)$ be the second-order action correction of (17), and define the normalized Hochschild one-cochain

$$\mathbf{e}(a \otimes b^{\text{op}}) = U(a)R(b) + L(a)W(b) + X_\xi(a)Y_\xi(b).$$

It is the coefficient of v^2 in the enveloping-algebra representation associated with the second-order action deformation.

Exact cochain calculation gives the stronger identities

$$(22) \quad \mathbf{a} \smile \mathbf{a} = 0, \quad [\mathbf{a}, \mathbf{x}] = 0, \quad d_\mu \mathbf{e} = -\mathbf{x} \smile \mathbf{x}, \quad [\mathbf{a}, \mathbf{e}] = -2\mathbf{x} \smile \mathbf{x}.$$

In particular, the vanishing of the u^2 and uv quadratic sources occurs already at the intrinsic cochain level; a contraction cannot insert arbitrary tangent-valued corrections into those zero sources.

Put

$$\mathbf{s} = [\mathbf{x}, \mathbf{e}] \in C^2(S, E_\mu).$$

Then $d_\mu \mathbf{s} = 0$. We next choose a specific constrained one-cochain $\mathbf{k}$. Let E_{ij} denote the standard matrix unit in $M_3(\mathbb{C})$. On the two factor axes set

$$\begin{aligned} X_{\mathbf{k}}(\epsilon) &= -\frac{1}{2}E_{32}, & X_{\mathbf{k}}(\epsilon^2) &= -E_{33}, & X_{\mathbf{k}}(\epsilon^3) &= 0, \\ Y_{\mathbf{k}}(\epsilon) &= \frac{1}{2}E_{32}, & Y_{\mathbf{k}}(\epsilon^2) &= E_{33}, & Y_{\mathbf{k}}(\epsilon^3) &= 0, \end{aligned}$$

and set

$$\mathbf{k}(\epsilon^i \otimes (\epsilon^j)^{\text{op}}) = 0 \quad (i, j \geq 1).$$

The factor-axis pair is a weak B -compatible perturbation, hence $\mathbf{k} \in \mathfrak{g}_{B,p}^1$. Define

$$\widehat{\zeta} := \mathbf{s} + d_\mu \mathbf{k}.$$

Then $\widehat{\zeta}$ is closed and represents the same cubic Hochschild class as $\mathbf{s}$. Moreover,

$$(23) \quad \Lambda(\Psi^2(\widehat{\zeta})) = 8,$$

so $[\widehat{\zeta}] \neq 0$ in $H^2(\mathfrak{g}_{B,p})$.

Definition 12.17 (Acyclic-intersection criterion). *Let $(C^\bullet, d)$ be a cochain complex and $W \subset C^n$ a finite-dimensional subspace of source cochains that one wants a contraction projection to annihilate. We say that W is acyclicly placeable if*

$$W \cap Z^n \subseteq B^n.$$

Equivalently, the image of $W \cap Z^n$ in $H^n(C)$ is zero.

Lemma 12.18 (Placement in a contraction complement). *If $W \subset C^n$ is acyclicly placeable, then one can choose a standard vector-space contraction of $C^\bullet$ onto its cohomology for which the degree- n projection vanishes on W .*

Proof. Choose splittings

$$Z^n = B^n \oplus H_{\text{rep}}^n.$$

Because $W \cap Z^n \subseteq B^n$, the image of W in C^n/Z^n has the same dimension as the image of $W/(W \cap B^n)$. Hence a complement C_{ac}^n of Z^n may be chosen to contain a complement of $W \cap B^n$ in W . Then

$$C^n = B^n \oplus H_{\text{rep}}^n \oplus C_{\text{ac}}^n$$

has projection onto H_{rep}^n equal to zero on W . Extending such splittings degree-wise gives the usual contraction. $\square$

Theorem 12.19 (Intrinsic quartic vanishing contraction). *For the cubic reconstruction of Section 12, there exists a contraction of the intrinsic constrained Hochschild DGLA onto cohomology such that*

$$\boxed{\ell_4|_{\text{Sym}^4 H^1} = 0.}$$

The same contraction retains the nonzero cubic obstruction, which may be normalized as

$$\ell_3(u[\mathbf{a}] + v[\mathbf{x}], \dots) \sim v^3[\widehat{\zeta}].$$

Consequently the nonzero presentation class $\nu_4(\alpha, \xi)$ of Theorem 12.12 does not give a quartic obstruction that is independent of the contraction in the intrinsic constrained Hochschild DGLA.

Proof. First note that the two boundaries

$$\mathbf{x} \smile \mathbf{x}, \quad d_\mu \mathbf{k}$$

are linearly independent. Hence the degree-one splitting may be chosen so that both $\mathbf{e}$ and $\mathbf{k}$ lie in its acyclic complement, with

$$h(\mathbf{x} \smile \mathbf{x}) = -\mathbf{e}.$$

Choose the cubic cohomology representative to be

$$i[\widehat{\zeta}] = \widehat{\zeta} = \mathbf{s} + d_\mu \mathbf{k}.$$

Then the standard contraction identity gives

$$h(\mathbf{s}) = -\mathbf{k}.$$

Using (22), the transferred Maurer–Cartan section through cubic order can therefore be taken as

$$F_1 = u\mathbf{a} + v\mathbf{x}, \quad F_2 = v^2\mathbf{e}, \quad F_3 = -2uv^2\mathbf{e} + v^3\mathbf{k}.$$

Its cubic residual is exactly $v^3\widehat{\zeta}$.

The quartic source has no u^4 or u^3v term. Its remaining coefficients are

$$\begin{aligned} Y_{22} &= [\mathbf{a}, -2\mathbf{e}] = 4\mathbf{x} \smile \mathbf{x}, \\ Y_{13} &= [\mathbf{a}, \mathbf{k}] - 2[\mathbf{x}, \mathbf{e}], \\ Y_{04} &= [\mathbf{x}, \mathbf{k}] + \mathbf{e} \smile \mathbf{e}. \end{aligned}$$

The first is a boundary. The other two are not closed. More strongly, their degree-three differentials are linearly independent. Evaluating them on

$$(1 \otimes \epsilon^{\text{op}})^{\otimes 3}$$

and taking the $(1, 2)$ and $(1, 3)$ matrix entries gives the exact 2×2 minor

$$\begin{pmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad \det = \frac{1}{4} \neq 0.$$

Therefore

$$\text{span}\{Y_{22}, Y_{13}, Y_{04}\} \cap Z^2 \subseteq B^2.$$

By Lemma 12.18, the degree-two contraction projection can be chosen to annihilate all three quartic source coefficients. This gives $\ell_4|_{\text{Sym}^4 H^1} = 0$.

Finally, the cubic class remains nonzero by (23). Thus the vanishing of the intrinsic quartic transfer is not obtained by killing the cubic obstruction. $\square$

Proposition 12.20 (Why the presentation class survives). *For the contraction above, the quartic Hochschild cochains satisfy*

$$\Gamma(\Psi^2(Y_{13})) = -16, \quad \Gamma(\Psi^2(Y_{04})) = -\frac{7}{2}.$$

Thus the two intrinsic quartic sources can lie in an acyclic Hochschild complement even though their presentation images are detected by the sparse presentation certificate.

Proof. Both identities are exact substitutions. There is no contradiction with Theorem 12.19: the intrinsic sources Y_{13}, Y_{04} are not Hochschild cocycles, whereas the presentation DGLA has no degree-three term. Since

$$\Psi^3 = 0,$$

the degree-three differentials that make these directions acyclic are forgotten by the finite relation model. Their images therefore become legitimate degree-two presentation classes even though the source cochains may be placed entirely in the intrinsic acyclic complement. $\square$

Corollary 12.21 (Failure of a strict intrinsic lift of ν_4). *The presentation class $\nu_4(\alpha, \xi)$ has no lift whose nonvanishing is a contraction-independent obstruction to*

$$\ell_4|_{\text{Sym}^4 H^1} = 0$$

in the constrained Hochschild DGLA. In particular, the presentation quartic no-go is a truncation-sensitive normal-form phenomenon rather than an intrinsic arity-four obstruction of the full Hochschild deformation complex.

Remark 12.22 (The correct intrinsic arity-four viewpoint). *For the full Hochschild complex the relevant chain-level datum is not an arbitrary quotient of C^2 by presentation relations. It is the harmonic intersection*

$$\frac{W_4 \cap Z^2}{W_4 \cap B^2},$$

where W_4 is the span of the quartic source cochains produced by the chosen lower-order transfer. Vanishing of this quotient is precisely what allows W_4 to be placed in the acyclic complement. In the contraction constructed above this quotient is zero.

12.4. One contraction through arity seven. The extension to arities five, six and seven uses the same test: the differential must be injective on the joint space of new nonclosed sources and previously fixed complement vectors. Together with independence of the degree-one lifts, Lemma 12.18 then places all these vectors in one contraction. Testing the new arity in isolation would not establish compatibility with earlier choices.

TABLE 2. Successive joint-source certificates for one intrinsic contraction.

New arity	Degree-one lifts	Joint differential rank	Nonzero minor
5	4	5	1/18
6	7	9	-1/486
7	11	14	11/34992

The complete cochains, source expansions and minors are retained in Appendix B, where Theorems B.1–B.5 prove the successive extensions. Thus one contraction preserves the cubic class and kills arities 4 through 7. The next section continues this same finite-data construction. Later low-rail formulas use the cochains $\mathbf{p}, \mathbf{q}, \mathbf{r}, \mathbf{w}, \mathbf{t}$ defined in that appendix.

13. FINITE FEEDBACK BEYOND ARITY SEVEN

The intrinsic transfer calculation of the preceding section can be reorganized as a finite-feedback problem. The point of this reorganization is not merely computational. It separates three structures that behave differently at higher order: the harmonic obstruction sector, the finite controller box in degree one, and the actuator directions by which the controller moves future Maurer–Cartan sources out of the closed sector.

Throughout this section we remain in the cubic example of Section 12. Write

$$S = A^e \cong \mathbb{C}[x, y]/(x^4, y^4), \quad E = \text{End}(V) \cong M_3(\mathbb{C}),$$

and denote by $\alpha, \xi \in H^1$ the tangent generators used above. The low-order homotopy cochains $\mathbf{e}, \mathbf{k}, \mathbf{p}, \dots$ are those fixed by the intrinsic contraction.

The two completion endpoints are compared in Figure 9.

A shared cubic germ and saved partial homotopy admit different completed transfers.

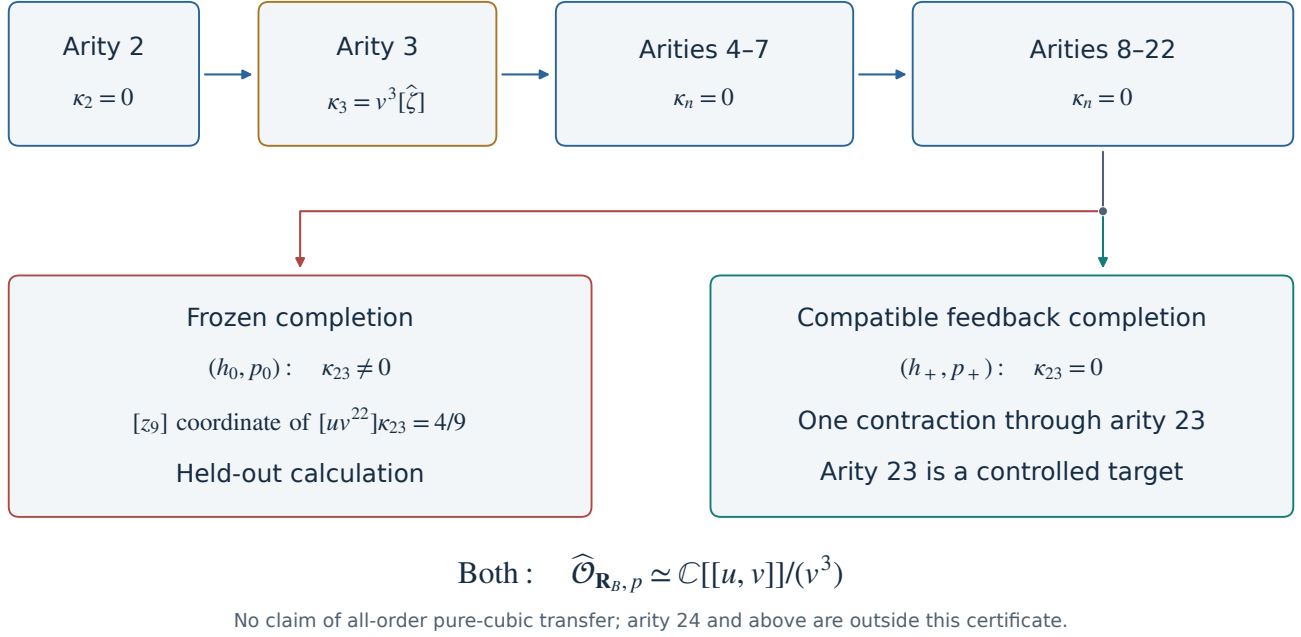


FIGURE 9. Arity comparison for two completions of the saved partial homotopy. The frozen completion has a held-out nonzero arity-23 coefficient (Proposition 13.3); a different compatible completion controls arity 23 to zero (Theorem 13.6). Both have the same cubic formal germ. Neither result asserts vanishing at arity 24 or above.

13.1. A split pure-mixed reservoir. Let

$$M_{\text{mix}} = \bigoplus_{1 \leq i, j \leq 3} E \delta_{ij} \subset \mathfrak{g}_{B,p}^1$$

be the normalized one-cochains that vanish on both factor axes and are supported on mixed monomials $x^i y^j$. For such a cochain f , the Hochschild differential satisfies

$$(df)(x^i, y^j) = -f(x^i y^j).$$

Consequently the restriction map

$$\rho : \mathfrak{g}_{B,p}^2 \longrightarrow M_{\text{mix}}, \quad (\rho\Phi)(x^i y^j) = -\Phi(x^i, y^j),$$

satisfies

$$(24) \quad \boxed{\rho d|_{M_{\text{mix}}} = \text{id.}}$$

Thus the mixed degree-one directions form an explicit split contractible reservoir. This proves compatibility of individual mixed lifts, but it does not by itself imply that all recursively generated degree-two sources lie in one acyclic complement. The additional condition required at arity n is

$$W_{\leq n}^{\text{nc}} \cap Z^2 = 0,$$

where $W_{\leq n}^{\text{nc}}$ denotes the span of the nonclosed source history through that arity.

13.2. Independent reconstruction through arity twenty-two. The surviving v0.33 file contains a partial homotopy on a source space K_{old} of dimension 100. Its name emphasizes low rails, but the stored cochains contain more information: the complete degree-one recursion through arity 17 can be regenerated from this partial map and the boundary splitting. No missing historical harmonic detector is needed for this statement. We work over $\mathbb{Q}$ and extend scalars to $\mathbb{C}$ afterwards.

Let $B^2 = dg_{B,p}^1$. Choose a complement C_{ac}^1 to $\text{span}\{\mathbf{a}, \mathbf{x}\}$ by taking, in order, the existing 35 controller vectors, the remaining weak-axis basis vectors, and the mixed matrix units, retaining a vector exactly when its differential increases rank. This gives $\dim C_{\text{ac}}^1 = \dim B^2 = 88$ and fixes $(d|_{C_{\text{ac}}^1})^{-1}$. For every archived pair $(q, h_{\text{old}}q)$ form the corrected source

$$w_q = q - dh_{\text{old}}q.$$

The span W_{old} of these corrected sources has

$$\dim W_{\text{old}} = \text{rank}(d|_{W_{\text{old}}}) = 99.$$

Thus it can be contained in a degree-two complement to the cycles.

Theorem 13.1 (Reconstructed single-contraction vanishing through arity 22). *There is a contraction of the constrained Hochschild complex retaining every archived homotopy assignment and the representative $\widehat{\zeta} = [\mathbf{x}, \mathbf{e}] + dk$ for which*

$$\kappa_2 = 0, \quad \kappa_3(u, v) = v^3[\widehat{\zeta}], \quad \kappa_n = 0 \quad (4 \leq n \leq 22).$$

Equivalently, up to the usual factorial normalization, all operations $\ell_n|_{\text{Sym}^n H^1}$ vanish in this range except the nonzero cubic operation. A common corrected source history W_{22} satisfies

$$\dim W_{22} = \text{rank}(d|_{W_{22}}) = 174, \quad \dim(B^2 + \mathbb{Q}\widehat{\zeta} + W_{22}) = 263.$$

This is an independently chosen completion, not an identification with the historical fixed-detector controller.

Proof. For $F_1 = u\mathbf{a} + v\mathbf{x}$ and the previously displayed $F_2, \dots, F_6$, regenerate every coefficient of

$$R_n = \frac{1}{2} \sum_{i=1}^{n-1} [F_i, F_{n-i}].$$

The certificate specifies every subsequent coefficient $f_{n,a,b}$ of F_n . For each coefficient $q_{n,a,b}$ of R_n , it verifies

$$q_{n,a,b} + df_{n,a,b} \in W_{22},$$

except at $(n, a, b) = (3, 0, 3)$, where the left side is $\widehat{\zeta}$. The exact rank calculations give $d|_{W_{22}}$ injective and $\widehat{\zeta} \notin B^2$. Hence $B^2 \oplus \mathbb{Q}\widehat{\zeta} \oplus W_{22}$ is a direct sum. Extend W_{22} to a complement of Z^2 , extend $\widehat{\zeta}$ to representatives of H^2 , and define h_2 to be the inverse of d on B^2 and zero on the other two summands. The same construction in the adjacent degrees extends these data to a standard contraction. Its projection p_2 annihilates B^2 and W_{22} , so $-h_2 R_n = F_n$ and $p_2 R_n = \kappa_n$ give precisely the asserted values. This is the rooted-tree transfer recursion [11, Chapter 10]; no truncation of $d : C^2 \rightarrow C^3$ is made.

The reconstruction uses corrected sources throughout. When dq is independent of the accumulated differential history, a new specified lift f is chosen and $q + df$ is adjoined. When dq is dependent, the resulting closed residual is reduced modulo the full 88-dimensional boundary space. All such residuals vanish in the stated range. The table records cumulative ranks with the saved history already included; they are not the ranks of the shorter prefix alone.

arity	7–17	18	19	20	21	22
cumulative differential rank	99	112	126	141	157	174
new independent directions	0	13	14	15	16	17

The second verifier recomputes d_2 using matrix-unit multiplication, verifies these ranks over $\mathbb{Q}$ by a separate elimination implementation, and replays every polynomial source under one fixed pair (h_2, p_2) . $\square$

Proposition 13.2 (The full degree-two cohomology). *For the constrained cubic complex,*

$$\dim C^2 = 2025, \quad \text{rank } d_2 = 1895, \quad \dim Z^2 = 130, \quad \dim H^2 = 42.$$

Proof. There are 15 normalized algebra inputs and 9 matrix entries, so the matrix of d_2 has size 30375×2025 . The matrix-unit Hochschild formula gives 29430 nonzero entries. Exact rational elimination supplies a 130-vector kernel basis, checked by multiplication by d_2 , and rank 1895. Subtracting the independently verified boundary dimension 88 gives 42. The verifier constructs this full matrix from the algebra action, not from the source-history ranks. $\square$

The two historical harmonic detector coordinates concern a restricted quadratic source space, not the full 42-dimensional H^2 . Nor should this dimension be confused with the one-dimensional effective relation space of the cubic formal germ.

A fully specified degree-two projection. Let $f_1, \dots, f_{88}$ be the chosen boundary lifts, let $z_1 = \widehat{\zeta}, z_2, \dots, z_{42}$ be the selected full cohomology representatives, and let $w_1, \dots, w_{174}$ be the corrected history basis. In normalized matrix-unit coordinates form

$$M = [df_1 \cdots df_{88} \mid z_1 \cdots z_{42} \mid w_1 \cdots w_{174}] \in \mathbb{Q}^{2025 \times 304}.$$

Select the lexicographic pivot rows I of M and write $T = (M_I)^{-1}$. For any $q \in C^2$, put $x = Tq_I$. The full maps are

$$h_2(q) = \sum_{j=1}^{88} x_j f_j, \quad p_2(q) = \sum_{j=1}^{42} x_{88+j} [z_j].$$

The coordinate vectors e_k , $k \notin I$, together with the 174 history vectors form W^2 of dimension 1895. This is a direct complement to Z^2 : restricting a proposed dependence to rows I and multiplying by T kills every coefficient in M , and then every remaining coordinate coefficient. The file `complete_splitting.json` supplies all basis vectors, I , T , and all 2025 columns of h_2, p_2 . The identities $h_2(df_j) = f_j$, $p_2(df_j) = 0$, $h_2(z_j) = 0$, $p_2(z_j) = [z_j]$, and $h_2(W^2) = p_2(W^2) = 0$ are checked on the full bases.

Proposition 13.3 (Held-out nonvanishing for the frozen completion). *For this fully specified completion, $\kappa_{23} \neq 0$. In the ordered cohomology basis just specified,*

$$\boxed{[z_9]\text{-coordinate of } [uv^{22}]\kappa_{23} = \frac{4}{9}.$$

There are 18 nonzero harmonic monomial coefficients, with v -degrees $6, \dots, 23$. The four coefficients with v -degrees $2, 3, 4, 5$ still have zero projection, and $F_{23}^{(18,5)}$ agrees with the archived low-rail value.

Proof. The complete projection file was hashed and fixed before regenerating

$$R_{23} = \frac{1}{2} \sum_{i=1}^{22} [F_i, F_{23-i}].$$

Applying the stored p_2 to every coefficient gives the stated values; the script verifies the same file hash afterwards and makes no further homotopy or projection choices. The full source coefficients and rational projection coordinates are in `heldout_arity23.json`. The nonzero coordinate $4/9$ is an exact witness. $\square$

Thus arity 22 is the last consecutive vanishing arity after the cubic term for this particular completed contraction. This is not a contraction-independent arity-23 obstruction, and does not alter the formal germ $\mathbb{C}[[u, v]]/(v^3)$.

Remark 13.4 (What has and has not been reconstructed). *The new certificate preserves the archived partial map and the available four low rails; new completion data may use a larger controller than U_{16} and a different projection away from the old source space. Matching the ranks 126, 141, 157, 174 therefore does not recover the historical matrix entries, distinguished sensitivities, pivot determinants, or quotient-history flags. Those finer fixed-detector statements are excluded from this version's proof chain. There is no claim here that all ℓ_n with $n \geq 24$ vanish, nor that the fixed quintic spectral factor is contraction-independent.*

13.3. Compatible stationary feedback through arity twenty-three. We now choose a different completion, denoted (h_+, p_+) , and keep the preceding completion as (h_0, p_0) . All comparisons use the same 42 cohomology representatives. The construction below is new; matching historical matrix sizes does not identify its detectors, pivot minors, or control coefficients with the unavailable originals.

Let $U = U_{16} \subset C_{ac}^1$ be the 35-dimensional controller with ordered basis $U_1, \dots, U_{35}$ specified in the companion's `blocks.json`. These are actual normalized matrix-unit cochains, not formal control labels. Define the quadratic source space

$$\mathcal{T} = \text{span}\{[f, g] : f, g \in \text{span}(\mathbf{a}, \mathbf{x}, U)\}.$$

The 37 generators give 703 unordered bracket pairs. Exact computation gives

$$\begin{aligned} \dim \mathcal{T} &= 267, & \text{rank}(d_2|_{\mathcal{T}}) &= 264, & \dim(\mathcal{T} \cap B^2) &= 1, \\ \dim(\mathcal{T} \cap Z^2) &= 3, & \dim p_0(\mathcal{T}) &= 3, & \dim C_{\text{quad}} &= 2, & C_{\text{quad}} &:= p_0(\mathcal{T} \cap Z^2). \end{aligned}$$

In particular the third projected direction comes from nonclosed sources; the full cohomology has not become two-dimensional.

Lemma 13.5 (Changing the projection off the cycles). *Choose a retraction $r : p_0(\mathcal{T}) \rightarrow C_{\text{quad}}$. There exists $L : C^3 \rightarrow H^2$ such that*

$$p_+ = p_0 + Ld_2, \quad p_+|_{Z^2} = p_0|_{Z^2}, \quad p_+(\mathcal{T}) = C_{\text{quad}}.$$

If $p_0(q) = 0$ for $q \in \mathcal{T}$, then also $p_+(q) = 0$.

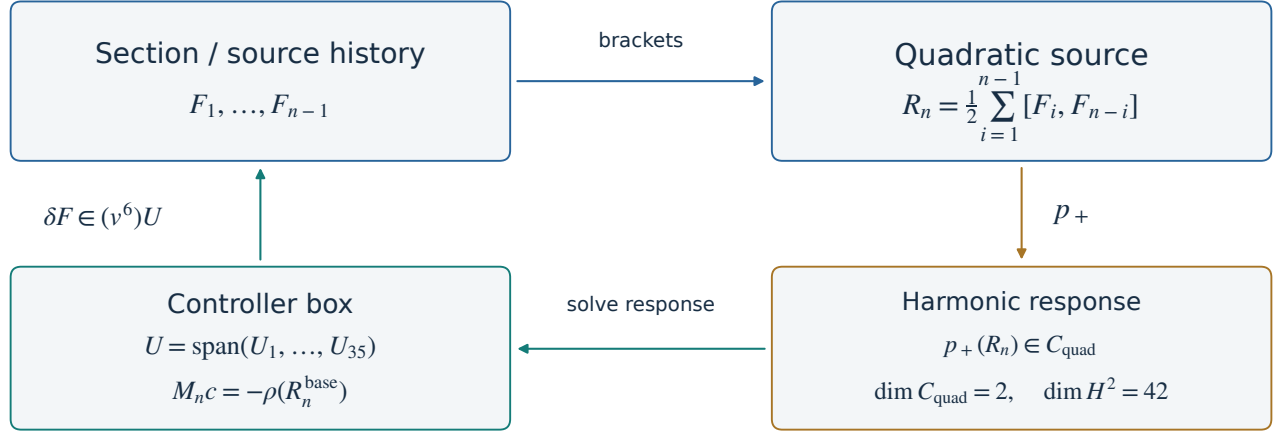
Proof. On $\mathcal{T}$ put $\delta = (r - I)p_0$. It vanishes on $\mathcal{T} \cap \ker d_2$, since r is the identity on C_{quad} . Thus $L(d_2q) = \delta(q)$ is well-defined on $d_2(\mathcal{T})$. Extend this linear map to C^3 . The displayed identities follow directly. Surjectivity onto C_{quad} follows already from the closed sources. $\square$

The certificate chooses the first independent extra image in an exact echelon basis and sends it to zero under r . It verifies that this retraction preserves the two-dimensional combined response of $[\mathbf{a}, U]$ and $[\mathbf{x}, U]$. The resulting p_+ is serialized on all 2025 columns before any new control solution is computed. Its values on all 100 saved sources are separately checked to be zero. No historical sensitivity is used to tune this choice.

Let $\rho : \mathcal{T} \rightarrow \mathbb{Q}^2$ be the coordinates of p_+ in the specified basis of C_{quad} . Put

$$A_j = \rho[\mathbf{a}, U_j], \quad B_j = \rho[\mathbf{x}, U_j].$$

The controller, harmonic response and acyclic history must fit one contraction.



$$\text{Actuation : } [a, U_j], [x, U_j] \quad (F_1 = ua + vx)$$

$$\rho_{\text{mix}} dM_{\text{mix}} = \text{id}, \quad W_{\leq n}^{\text{nc}} \cap Z^2 = 0$$

The split mixed reservoir supplies lifts; the separate history condition enforces compatibility.

FIGURE 10. The staged finite-feedback construction. A 35-dimensional controller changes high- v section coefficients and hence future quadratic sources. The response sector C_{quad} has dimension two, while full H^2 has dimension 42. The split mixed reservoir (here denoted by ρ_{mix}) supplies lifts; the separate source-history condition makes them compatible with one contraction.

For target arity n , a control in the preceding section coefficient is $c_{b,j} u^{n-1-b} v^b U_j$, where $6 \leq b \leq n-1$. Its contribution to R_n is its bracket with $F_1 = ua + vx$. Consequently the entire feedback matrix is

$$(25) \quad (M_n)_{(d,k),(b,j)} = \delta_{d,b}(A_j)_k + \delta_{d,b+1}(B_j)_k, \quad \begin{array}{l} 6 \leq d \leq n, \quad k = 1, 2, \\ 6 \leq b \leq n-1, \quad 1 \leq j \leq 35. \end{array}$$

The scalar sector coordinates and basis indices in the JSON files are zero-based; the formula above uses one-based indices.

The corresponding structures are illustrated in Figure 10.

Theorem 13.6 (One compatible contraction through arity 23). *For the fixed cubic complex there is one contraction (h_+, p_+) retaining the original 100 partial-homotopy assignments and the full cohomology representatives, such that*

$$\boxed{\kappa_2 = 0, \quad \kappa_3 = v^3[\hat{\zeta}], \quad \kappa_n = 0 \quad (4 \leq n \leq 23).}$$

Its $F_2, \dots, F_{22}$ lie in U . The five feedback systems below have the stated exact coefficient and augmented ranks.

target arity	rows	columns	rank / augmented rank
19	28	455	28 / 28
20	30	490	30 / 30
21	32	525	32 / 32
22	34	560	34 / 34
23	36	595	36 / 36

Here arity 23 is controlled, not a held-out prediction. No assertion about arity 24 or all-order pure-cubic transfer is made.

Proof. Keep $F_1, \dots, F_{17}$ from the saved partial map. At each step $18 \leq m \leq 22$, reduce every coefficient of R_m modulo the already specified homotopy domain. The saved reductions show that the channels with v -degrees $6, \dots, m$ are independent modulo that domain; their homotopy values can therefore be chosen freely in U . Dependent channels are assigned their uniquely forced values. First set the free values to zero and compute the complete next source R_{m+1}^{base} from all preceding coefficients.

Solve

$$M_{m+1}c = -\rho(R_{m+1}^{\text{base}}).$$

Each file `feedback_n.json` contains every matrix entry, the actual source before control, the right-hand side with this minus sign, and a rational solution. A square pivot minor P_n is accompanied by its inverse; exact multiplication checks $P_n P_n^{-1} = I$ and recomputes its determinant. This proves full row rank. Exact substitution $M_n c = \text{rhs}$ proves compatibility with the actual source, rather than only a rank claim. All source coefficients belong to $\mathcal{T}$, so the two sector equations are the full harmonic equations, by Lemma 13.5. The coefficients of v -degrees below six are fixed by the preserved low rails and have zero projection; they too are replayed, not omitted.

Prescribe $h_+(q) = -f$ for every source/section pair q, f , while retaining h_{old} on the saved domain and the inverse on the full boundary space. Exact reduction verifies all linear dependencies of these assignments. The cumulative differential ranks, with the old history included, are

$$99, 112, 126, 141, 157, 174$$

through arities 17, 18, 19, 20, 21, 22 respectively. On this entire domain the difference from h_0 vanishes on closed combinations, and hence factors through d_2 . Extending that factor linearly gives $h_+ = h_0 + L_h d_2$ on all of C^2 , with image in the fixed 88-dimensional C_{ac}^1 . The full-column certificate checks

$$h_+ d_1|_{C_{\text{ac}}^1} = I, \quad p_+ d_1 = 0, \quad h_+ i_2 = 0, \quad p_+ i_2 = I.$$

Since $Z^2 = B^2 \oplus i_2 H^2$, the map $Q = I - d_1 h_+ - i_2 p_+$ has image

$$W^2 = \ker h_+ \cap \ker p_+,$$

is the identity there, and kills Z^2 . Thus $C^2 = B^2 \oplus i_2 H^2 \oplus W^2$ and $\dim W^2 = 1895$. These splittings extend in adjacent degrees to a standard contraction.

Finally, the independent consumer recomputes every R_n from matrix-unit cup products and checks $F_n = -h_+ R_n$, $\kappa_n = p_+ R_n$ through 23. This includes the globally determined F_{23} , for which no U -membership assertion is needed. Hence all values come from one contraction, not from unrelated arity-by-arity projections. $\square$

Remark 13.7 (Why the trajectory had to be rebuilt). *Keeping p_0 and the earlier unrestricted trajectory while resetting its F_{21} high- v channels yields a U -restricted arity-22 system of coefficient rank 34 but augmented rank 35 in the full harmonic space. The corresponding 88-direction*

system has ranks 67/67. Thus the successful new 34-row system cannot be attached retroactively to that old trajectory. The separate countercheck reproduces this incompatibility.

Corollary 13.8 (The finite endpoint depends on the completion). *The fixed cubic data and the saved partial-homotopy assignments do not determine whether κ_{23} vanishes.*

Proof. Proposition 13.3 gives a nonzero value for (h_0, p_0) ; Theorem 13.6 gives zero for (h_+, p_+) , with the same saved assignments. Both have the same cubic formal germ. $\square$

14. AUTONOMOUS LOW-RAIL TRANSFER STATE

The finite feedback construction and the all-orders low-rail theorem have different scopes. The former controls the full harmonic source through arity 23. The latter fixes the finite low-rail homotopy data and studies the resulting autonomous triangular subsystem for all temporal indices.

Proposition 14.1 (Low-rail independence). *For a fixed homotopy, perturbations supported at v -degree at least six do not change the recursively generated coefficients of v -degrees 2, 3, 4, 5.*

Proof. The bracket adds polynomial degrees and the linear homotopy acts only on cochain coefficients. The ideal (v^6) is therefore preserved by every bracket contribution containing such a perturbation and by the homotopy. Induction on arity proves equality of the recursions modulo (v^6) . $\square$

This is an independence statement for the four low rails. It does not claim that indirect propagation through the v^6 rail is zero. In particular no unrecovered historical distinguished-detector sequence is needed below.

14.1. Autonomous low rails. The actual memory lies below v -degree six. Define

$$(26) \quad \boxed{x_b(k) = F_{k+b}^{(k,b)}, \quad b = 2, 3, 4, 5.}$$

These cochain-valued sequences satisfy exact triangular quadratic recursions. With h denoting the fixed intrinsic homotopy,

$$(27) \quad x_2(k) = -h[\alpha, x_2(k-1)], \quad x_2(k) = (-2)^k \mathbf{e}.$$

Next,

$$(28) \quad x_3(k) = -h([\alpha, x_3(k-1)] + [\xi, x_2(k)]).$$

The v^4 rail obeys

$$(29) \quad \begin{aligned} x_4(k) = -h \Big(& [\alpha, x_4(k-1)] + [\xi, x_3(k)] \\ & + \frac{1}{2} \sum_{a+b=k} [x_2(a), x_2(b)] \Big), \end{aligned}$$

and the v^5 rail obeys

$$(30) \quad \begin{aligned} x_5(k) = -h \Big(& [\alpha, x_5(k-1)] + [\xi, x_4(k)] \\ & + \frac{1}{2} \sum_{a+b=k} ([x_2(a), x_3(b)] + [x_3(b), x_2(a)]) \Big). \end{aligned}$$

In these recurrences $k \geq 1$, and every convolution index is nonnegative. The initial coefficients are

$$x_2(0) = \mathbf{e}, \quad x_3(0) = \mathbf{k}, \quad x_4(0) = -\mathbf{q}, \quad x_5(0) = -\mathbf{t},$$

as fixed by the section coefficients above. The polynomial $q(S)$ below is not the cochain $\mathbf{q}$. Available transitions and the two later frozen checks are replayed from these initial data.

The cochain-mode ranks on the exact trajectory are

$$(31) \quad \boxed{\dim W_2 = 1, \quad \dim W_3 = 7, \quad \dim W_4 = 7, \quad \dim W_5 = 13,}$$

where $W_b = \text{span}\{x_b(k)\}$ over the computed range.

14.2. Frozen transfer law and two out-of-sample tests. Define

$$q(\lambda) = \lambda^5 - 4\lambda^4 + 12\lambda^3 - 32\lambda^2 + 80\lambda - 192.$$

On the initial exact trajectory the four low rails admit the annihilators

$$\begin{aligned} p_2(\lambda) &= \lambda + 2, & p_3(\lambda) &= (\lambda + 2)^2 q(\lambda), \\ p_4(\lambda) &= (\lambda + 2)^3 q(\lambda), & p_5(\lambda) &= (\lambda + 2)^4 q(\lambda)^2. \end{aligned}$$

For

$$Y_k = (x_2(k), x_3(k), x_4(k), x_5(k)),$$

the first constant scalar recurrence on the fitting window has order fourteen:

$$(32) \quad \boxed{\begin{aligned} Y_{k+14} &= -589824Y_k - 688128Y_{k+1} - 200704Y_{k+2} \\ &\quad + 1536Y_{k+7} + 896Y_{k+8}. \end{aligned}}$$

Its characteristic polynomial is

$$(33) \quad \boxed{p(\lambda) = (\lambda + 2)^4 q(\lambda)^2.}$$

Equivalently,

$$p(\lambda) = \lambda^{14} - 896\lambda^8 - 1536\lambda^7 + 200704\lambda^2 + 688128\lambda + 589824.$$

Let M_{low} be the 14×14 companion matrix of (32). The matrix was frozen before computing further arities. It predicted Y_{17} and Y_{18} exactly. The first prediction was then checked by an independent arity-twenty-two quadratic recursion, and the second by an independent arity-twenty-three low-rail recursion. In both cases every cochain coordinate agreed. Thus the finite realization passed two genuine out-of-sample tests without refitting.

The corresponding structures are illustrated in Figure 11.

15. TRIANGULAR SPECTRAL EXTENSION AND RESIDUAL MODULES

We now prove that the factorization (33) is an all-orders consequence of the fixed cubic contraction, rather than a finite-window coincidence.

Let S denote the forward temporal shift,

$$(Sx)(k) = x(k+1),$$

and set

$$E = S + 2.$$

Define the homogeneous operator

$$A = -h \operatorname{ad}_\alpha.$$

The v^3 recursion implies that

$$g_3 := Ex_3$$

For fixed h , $x_b(k)$ is the coefficient of $u^k v^b$ in F_{k+b} , for $b = 2, 3, 4, 5$.

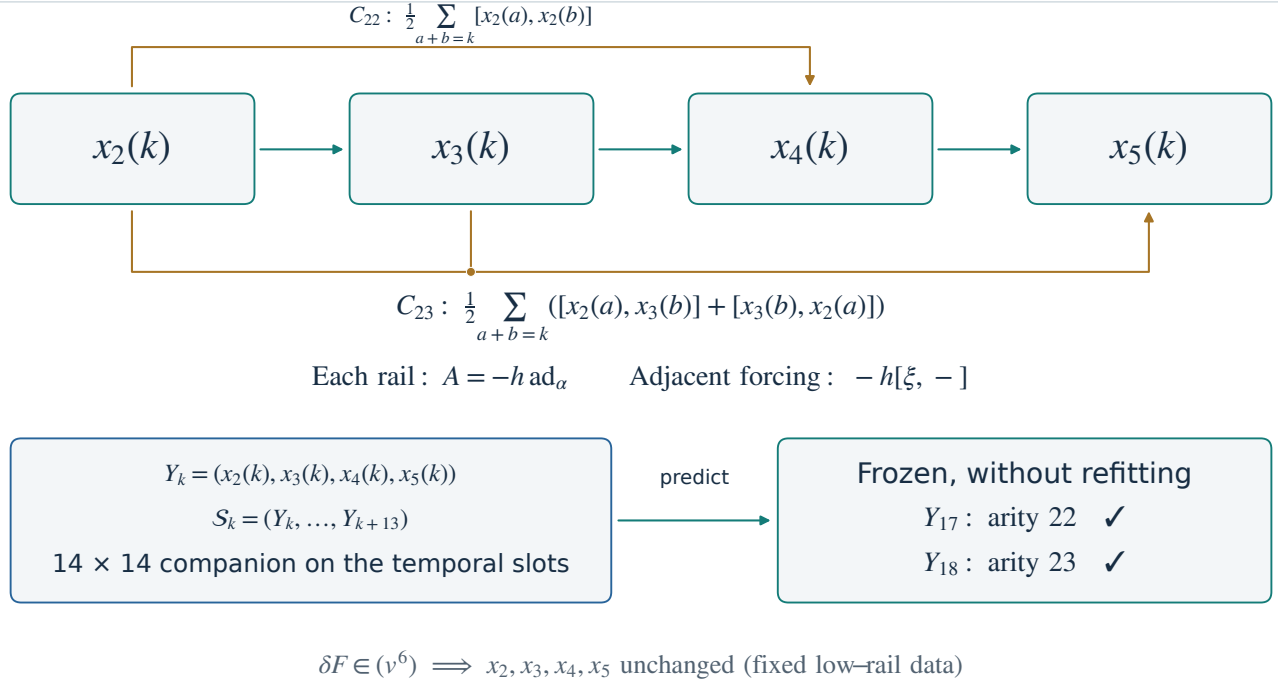


FIGURE 11. The four autonomous low rails. Adjacent arrows encode the $[\xi, -]$ forcing; the Cauchy terms C_{22} and C_{23} enter after applying $-h$. Each rail also has the homogeneous update $A = -h \operatorname{ad}_\alpha$. The companion acts on 14 temporal slots for each cochain coordinate. The frozen predictions of Y_{17} and Y_{18} agree with independent recursions without refitting. Independence modulo (v^6) does not assert future- v^6 terminality.

is homogeneous:

$$Sg_3 = Ag_3.$$

The exact cyclic space generated by $g_3(0)$ has dimension six. In the basis

$$g_3(0), Ag_3(0), \dots, A^5 g_3(0),$$

the operator is

$$(34) \quad A_* = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 384 \\ 1 & 0 & 0 & 0 & 0 & 32 \\ 0 & 1 & 0 & 0 & 0 & -16 \\ 0 & 0 & 1 & 0 & 0 & 8 \\ 0 & 0 & 0 & 1 & 0 & -4 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix}.$$

Its characteristic and minimal polynomial are both

$$(35) \quad \boxed{m_A(\lambda) = (\lambda + 2)q(\lambda).}$$

Lemma 15.1 (Resonant Cauchy forcing). *Let $a(k) = \lambda_0^k a_0$, let B be a fixed bilinear map, and define*

$$C_b(k) = \sum_{i+j=k} B(a(i), b(j)).$$

If $E = S - \lambda_0$, then

$$(EC_b)(k) = B(a_0, b(k+1)).$$

Consequently, $P(S)b = 0$ implies $EP(S)C_b = 0$.

Proof. Expand $C_b(k+1)$ and separate the term with $i = 0$. For $i \geq 1$, use $a(i) = \lambda_0 a(i-1)$ to identify the remaining sum with $\lambda_0 C_b(k)$. The final assertion follows because temporal polynomials commute with the fixed bilinear operation after one input is frozen to the geometric mode. $\square$

Define the residual sequences

$$r_b = m_A(S)x_b, \quad b = 3, 4, 5,$$

and their cyclic temporal modules

$$\mathcal{R}_b = \mathbb{C}[S]r_b.$$

Theorem 15.2 (Residual Module Theorem). *For the fixed cubic contraction,*

$$\mathcal{R}_3 \simeq \mathbb{C}[S]/(S+2), \quad \mathcal{R}_4 \simeq \mathbb{C}[S]/(S+2)^2,$$

and

$$\mu_{\mathcal{R}_5}(\lambda) = (\lambda+2)^3 q(\lambda).$$

Consequently

$$p_5(\lambda) = (\lambda+2)^4 q(\lambda)^2$$

annihilates the v^5 rail for all temporal indices for which the autonomous triangular subsystem is defined.

Proof. Since $g_3 = Ex_3$ is a cyclic homogeneous A -orbit with minimal polynomial $m_A = Eq$, one has

$$r_3 = q(S)g_3, \quad Er_3 = m_A(S)g_3 = 0.$$

If $r_3 = 0$, the degree-five polynomial q would annihilate the six-dimensional cyclic orbit g_3 , contradicting (35). Hence $\mathcal{R}_3 \simeq \mathbb{C}[S]/E$.

Put $p_3 = Em_A = E^2q$ and $y_4 = p_3(S)x_4 = Er_4$. The term $[\xi, x_3]$ is killed by p_3 , while Lemma 15.1 gives $E^2C_{22} = 0$ for the $x_2 * x_2$ convolution. Thus y_4 is homogeneous. Exact evaluation gives a nonzero eigenvector with

$$Ay_4(0) = -2y_4(0),$$

so $Ey_4 = 0$ but $y_4 \neq 0$. Therefore $E^2r_4 = 0$ and $Er_4 \neq 0$, proving the second isomorphism.

Now put $p_4 = Ep_3 = E^3q$ and $y_5 = p_4(S)x_5$. The term $[\xi, x_4]$ is killed by p_4 , and the resonant Cauchy lemma gives $p_4C_{23} = 0$. Hence y_5 is homogeneous. Its first six orbit vectors are cyclic and carry exactly the same matrix (34), so $\mu_{y_5} = m_A$. It follows that $m_A p_4$ annihilates x_5 and hence p_4 annihilates $r_5 = m_A x_5$. Finally the eight cochains $r_5(0), \dots, r_5(7)$ are linearly independent; an exact 8×8 rational minor has determinant

$$-1974127506397904857128197160960 \neq 0.$$

Since $\deg p_4 = 8$, no smaller polynomial annihilates r_5 . Thus $\mu_{\mathcal{R}_5} = p_4$. $\square$

Proposition 15.3 (Exact cyclic temporal modules). *Put $R = \mathbb{C}[S]$ and $\mathcal{X}_b = Rx_b$. Then*

$$\mathcal{X}_4 \simeq R/(p_4), \quad \mathcal{X}_5 \simeq R/(p_5).$$

Proof. The residual-module theorem gives the upper annihilators p_4 and p_5 . For the converse, the first eight shifts of x_4 , evaluated jointly at temporal indices 0, 1, are linearly independent. The corresponding fourteen shifts of x_5 are also independent. These are exact ranks 8 and 14 of the two-window block matrices

$$\begin{pmatrix} x_b(0) & x_b(1) & \cdots & x_b(d-1) \\ x_b(1) & x_b(2) & \cdots & x_b(d) \end{pmatrix}, \quad (b, d) = (4, 8), (5, 14),$$

after expanding cochains in matrix-unit coordinates. Both matrices are reconstructed by the companion; see Appendix A. Any temporal relation of degree less than d would give a dependence of these columns. Thus the upper annihilators are minimal. Apply the first isomorphism theorem to $R \rightarrow Rx_b$. $\square$

Theorem 15.4 (Triangular Spectral Extension Lemma). *Let V be finite-dimensional, let $A \in \text{End}(V)$, and let a temporal sequence satisfy*

$$Sx = Ax + f.$$

Suppose P is monic and $P(S)f = 0$. Set

$$y = P(S)x$$

and let

$$d_y(\lambda) = \mu(A|_{\mathbb{C}[A]y(0)}).$$

Then

$$\mu_x \mid P d_y.$$

If no polynomial of degree $< \deg P + \deg d_y$ annihilates x , then

$$\boxed{\mu_x = P d_y.}$$

In particular, if $y(0)$ is a nonzero λ_0 -eigenvector, the new multiplier is $\lambda - \lambda_0$; if $y(0)$ is cyclic for the full homogeneous A -module, the new multiplier is m_A .

Proof. Since $P(S)$ commutes with the constant operator A ,

$$(S - A)y = P(S)(S - A)x = P(S)f = 0.$$

Thus $y(k) = A^k y(0)$, so $d_y(S)y = 0$ and therefore $P(S)d_y(S)x = 0$. The equality statement follows from the stated lower-degree exclusion. $\square$

The cubic transitions are therefore

$$\boxed{p_3 = E m_A, \quad p_4 = E p_3, \quad p_5 = m_A p_4.}$$

They are, respectively, a full spectral extension, a simple resonant extension, and another full spectral extension.

The corresponding structures are illustrated in Figure 12.

Each forcing annihilator leaves a homogeneous landing, which supplies the next multiplier.

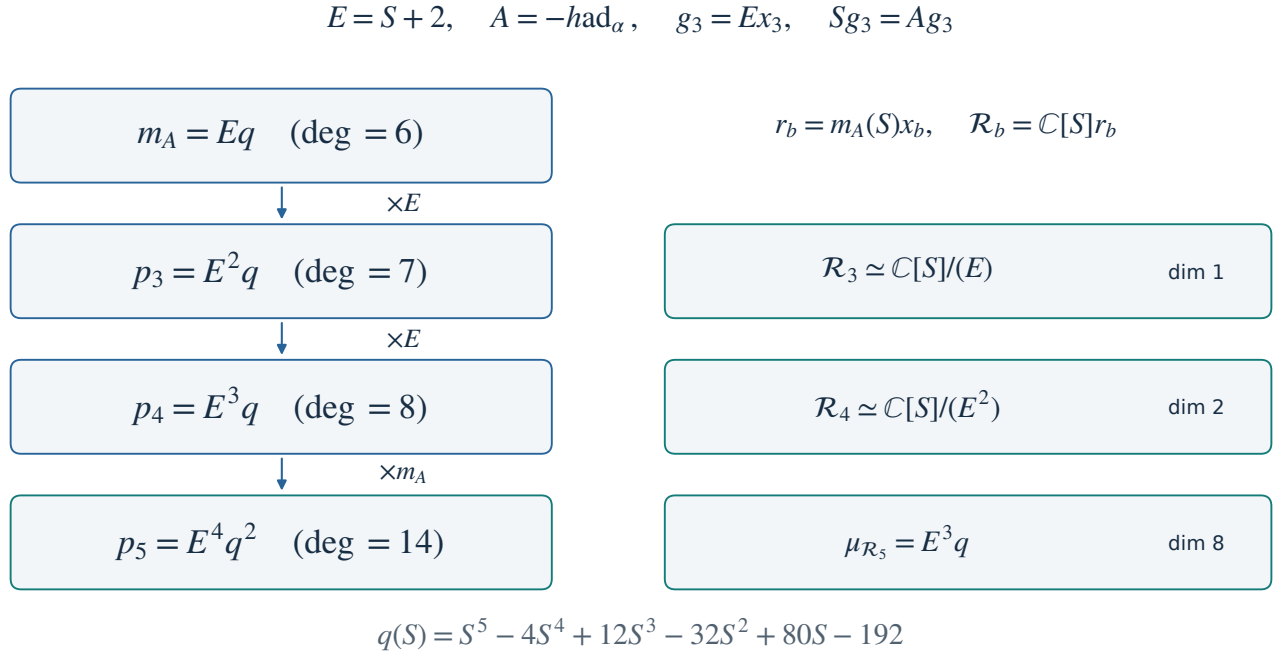


FIGURE 12. The triangular spectral extension and residual-module tower. With $E = S + 2$ and $m_A = Eq$, the successive multipliers are E, E, m_A . The residual modules have dimensions 1, 2, 8. The annihilators are exact all-orders conclusions for the fixed autonomous subsystem, not contraction-independent invariants.

16. RESIDUAL LANDING OPERATORS AND SPECTRAL SELECTION

The preceding lemma can be made predictive using only finite boundary data.

Proposition 16.1 (Finite-jet landing formula). *Let*

$$x_{k+1} = Ax_k + f_k, \quad P(z) = \sum_{r=0}^d p_r z^r, \quad P(S)f = 0.$$

Then the homogeneous landing vector

$$y_0 = P(S)x(0)$$

depends only on $x_0, f_0, \dots, f_{d-1}$ and is

$$y_0 = P(A)x_0 + \sum_{t=0}^{d-1} K_{P,t}(A)f_t,$$

where

$$K_{P,t}(z) = \sum_{r=t+1}^d p_r z^{r-1-t}.$$

Proof. Unroll the recurrence to

$$x_r = A^r x_0 + \sum_{t=0}^{r-1} A^{r-1-t} f_t$$

and substitute into $P(S)x(0) = \sum_r p_r x_r$. $\square$

The displayed formula computes a landing from an initial state and forcing jet. To specify the complete family independently of homogeneous ambiguity, use the fixed cyclic temporal modules of Proposition 15.3:

$$\mathcal{H}_4 = E^2 \mathcal{X}_4, \quad \mathcal{H}_5 = p_4 \mathcal{X}_5, \quad \mathcal{D}_b = \mathcal{X}_b / \mathcal{H}_b \quad (b = 4, 5).$$

Both $\mathcal{H}_b$ are cyclic R -modules with annihilator m_A . Identify their shift action with the homogeneous companion operator A_* using the generators $h_0 = E^2 x_4$ and $y_5 = p_4 x_5$ respectively. This is a specified cyclic-module identification, not an identification of arbitrary cochain jets. Define

$$\Lambda_4([u]) = p_3(S)u, \quad \Lambda_5([u]) = p_4(S)u.$$

Their values lie in $\mathcal{H}_4, \mathcal{H}_5$. Changing a lift in $\mathcal{H}_4$ contributes $p_3 E^2 = E p_4$, while changing one in $\mathcal{H}_5$ contributes $p_4^2 = E^2 p_5$. Both vanish in the respective $\mathcal{X}_b$, proving well-definedness.

Theorem 16.2 (Complete landing operators for x_4 and x_5). *For the fixed cubic contraction, let $R = \mathbb{C}[S]$ and $E = S + 2$.*

For x_4 ,

$$\mathcal{D}_4 \simeq R/E^2 R, \quad \dim \mathcal{D}_4 = 2.$$

In the domain basis $[x_4], [Sx_4]$, abbreviated $[1], [S]$, and target basis $h_0, Sh_0, \dots, S^5 h_0$, with shift matrix (34), the complete landing operator is

$$[\Lambda_4] = \begin{pmatrix} -192 & 384 \\ 80 & -160 \\ -32 & 64 \\ 12 & -24 \\ -4 & 8 \\ 1 & -2 \end{pmatrix}.$$

It satisfies

$$\boxed{\text{rank } \Lambda_4 = 1, \quad (A + 2I)\Lambda_4 = 0.}$$

Hence every nonzero admissible x_4 landing is structurally an E -extension.

For x_5 ,

$$\mathcal{D}_5 \simeq R/p_4 R, \quad \dim \mathcal{D}_5 = 8,$$

and, in the domain basis $[x_5], [Sx_5], \dots, [S^7 x_5]$, abbreviated $[1], [S], \dots, [S^7]$, and target basis $y_5, Sy_5, \dots, S^5 y_5$,

$$[\Lambda_5] = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 384 & 768 \\ 0 & 1 & 0 & 0 & 0 & 0 & 32 & 448 \\ 0 & 0 & 1 & 0 & 0 & 0 & -16 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 \end{pmatrix}.$$

Thus

$$\boxed{\text{rank } \Lambda_5 = 6,}$$

so Λ_5 is surjective onto the full homogeneous module. Moreover

$$\ker \Lambda_5 = m_A \mathcal{D}_5$$

is two-dimensional.

Proof. The domain identifications follow from $\mathcal{X}_4 = R/(E^2 m_A)$ and $\mathcal{X}_5 = R/(p_4 m_A)$. For Λ_4 the columns are $q(A_*)e_1$ and $A_* q(A_*)e_1 = -2q(A_*)e_1$, giving the nonzero rank-one eigenimage. For Λ_5 they are $e_1, A_* e_1, \dots, A_*^7 e_1$; the first six form the identity matrix. Its kernel consists exactly of classes divisible by m_A , of dimension $\deg p_4 - \deg m_A = 2$. $\square$

For $d = (d_0, \dots, d_7)$ write

$$r_d(S) = \sum_{j=0}^7 d_j S^j.$$

Then

$$\Lambda_5(d) = r_d(A)y_5.$$

The Krylov determinant satisfies

$$(36) \quad \boxed{\Delta_A(\Lambda_5(d)) = \det r_d(A) = \text{Res}(m_A, r_d).}$$

Since the datum $r_d = 1$ gives determinant one, this polynomial is nonzero; full m_A -extension is therefore Zariski-generic.

The corresponding structures are illustrated in Figure 13.

17. SPECTRAL AND EXTENSION STRATIFICATIONS

The complete landing geometry is controlled by elementary gcd data in the PID $\mathbb{C}[S]$.

Lemma 17.1 (GCD landing formula). *Let $\mathcal{H} = \mathbb{C}[S]/(m)$ with cyclic generator $y = [1]$. For $v = r(A)y$,*

$$\boxed{\text{Ann}_{\mathbb{C}[S]}(v) = \left(\frac{m}{\gcd(m, r)} \right)}.$$

Proof. Write $m = gm_1$ and $r = gr_1$ with $g = \gcd(m, r)$ and $\gcd(m_1, r_1) = 1$. Then $fv = 0$ iff $m \mid fr$, equivalently $m_1 \mid fr_1$, and hence $m_1 \mid f$. $\square$

For the cubic homogeneous polynomial $m_A = Eq$, exact symbolic calculation gives

$$q(-2) = -672 \neq 0, \quad \gcd(q, q') = 1, \quad \gcd(m_A, m'_A) = 1,$$

and q is irreducible over $\mathbb{Q}$. Over a splitting field write

$$q(S) = \prod_{i=1}^5 Q_i(S), \quad Q_i(S) = S - \beta_i.$$

Thus m_A has six distinct spectral roots.

Theorem 17.2 (Spectral landing stratification). *For $d \in \mathcal{D}_5$,*

$$\boxed{\mu_H(d) = \frac{m_A}{\gcd(m_A, r_d)}}.$$

Over a splitting field, let

$$Z(d) = \{\lambda \in \{-2, \beta_1, \dots, \beta_5\} : r_d(\lambda) = 0\}.$$

Then the locally closed strata

$$\mathcal{S}_Z = \{d : Z(d) = Z\}$$

$R = \mathbb{C}[S]$. Domains are forcing quotients; targets are specified homogeneous cyclic modules.

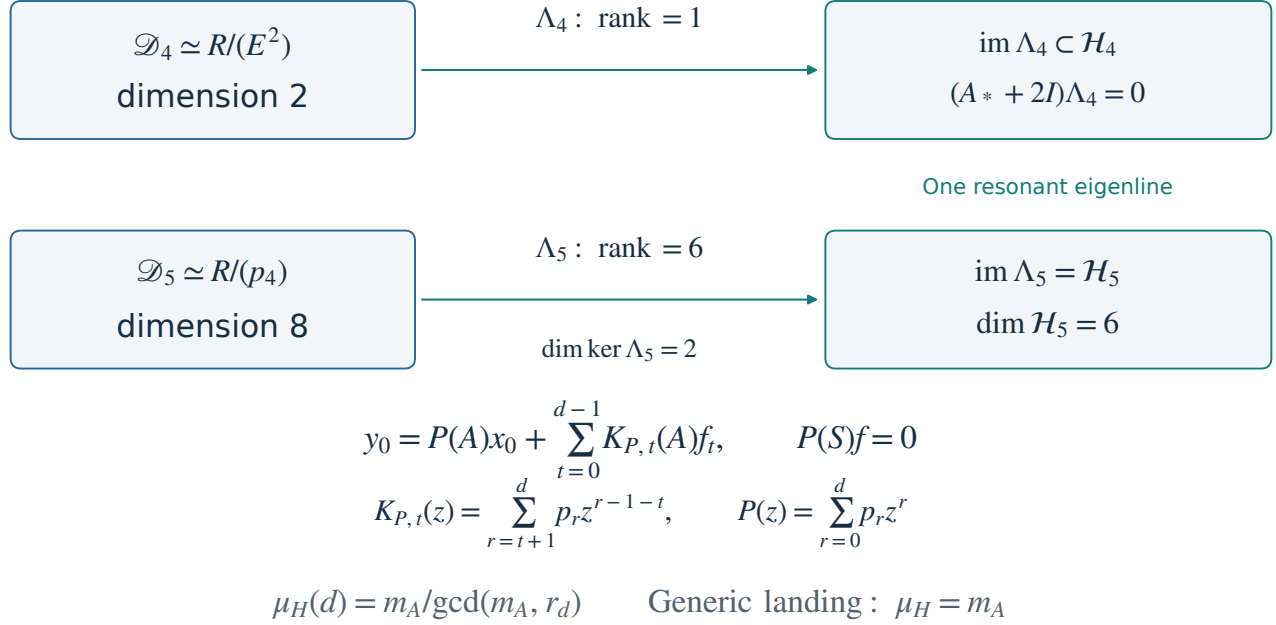


FIGURE 13. Complete landing operators and spectral selection. The rank-one map Λ_4 lands in the resonant eigenline; Λ_5 is onto the six-dimensional homogeneous module and has a two-dimensional kernel. The finite-jet formula assumes $P(S)f = 0$. In the fixed cyclic module, the surviving homogeneous annihilator is $m_A / \gcd(m_A, r_d)$.

are all nonempty, there are $2^6 = 64$ of them, and on $\mathcal{S}_Z$

$$\mu_H = \prod_{\lambda \notin Z} (S - \lambda).$$

Moreover

$$\dim \mathcal{S}_Z = 8 - |Z|$$

and

$$\overline{\mathcal{S}_Z} = \bigsqcup_{Z' \supseteq Z} \mathcal{S}_{Z'}.$$

Equation (36) factors over $\mathbb{Q}$ as

$$\text{Res}(m_A, r_d) = r_d(-2) \text{Res}(q, r_d).$$

Hence the rational exceptional divisor has a resonant linear component and an irreducible quintic norm component. Rational data can therefore have only the four homogeneous multipliers

$$m_A, \quad q, \quad E, \quad 1.$$

The forcing quotient retains higher resonant depth. Set

$$g_H = \gcd(m_A, r_d), \quad g_F = \gcd(p_4, r_d).$$

Theorem 17.3 (Two-axis forcing/spectral stratification). *Since $m_A \mid p_4$,*

$$g_H = \gcd(m_A, g_F).$$

Over a splitting field define

$$\nu(d) = \min\{3, \text{ord}_E r_d\}$$

and

$$Z(d) = \{i : r_d(\beta_i) = 0\}.$$

Then the common refinement is indexed by

$$(\nu, Z) \in \{0, 1, 2, 3\} \times 2^{\{1, \dots, 5\}},$$

with

$$g_F = E^\nu \prod_{i \in Z} Q_i, \quad g_H = E^{\min(1, \nu)} \prod_{i \in Z} Q_i.$$

There are 128 nonempty strata, with

$$\dim \mathcal{S}_{\nu, Z} = 8 - \nu - |Z|,$$

and closure order

$$(\nu, Z) \preceq (\nu', Z') \iff \nu \leq \nu', \quad Z \subseteq Z'.$$

The codimension generating polynomial is

$$(1 + t + t^2 + t^3)(1 + t)^5.$$

The two exceptional resultants have the same reduced support but different resonant multiplicity:

$$\begin{aligned} \text{Res}(m_A, r_d) &= r_d(-2) \text{Res}(q, r_d), \\ \text{Res}(p_4, r_d) &= r_d(-2)^3 \text{Res}(q, r_d). \end{aligned}$$

Thus forcing memory is a resonant jet refinement of reduced spectral support.

Finally introduce the canonical higher-lift invariant

$$g_X(d) = \gcd(p_5, r_d), \quad p_5 = E^4 q^2.$$

Here r_d is the unique representative of degree < 8 in $\mathcal{D}_5 \simeq R/(p_4)$. The chosen polynomial section is

$$X_{\text{can}}(d) := r_d(S)x_5 \in \mathcal{X}_5.$$

It is a $\mathbb{C}$ -linear section, not an R -linear splitting of the extension. While g_H and g_F do not depend on the polynomial representative, g_X describes this specified lift. In a cyclic module of annihilator p_5 , the annihilator of $r_d x_5$ is $p_5 / \gcd(p_5, r_d)$, which explains the full-lift formula below.

Theorem 17.4 (Three-level extension stratification). *For every canonical datum,*

$$g_H \mid g_F \mid g_X, \quad g_H = \gcd(m_A, g_X), \quad g_F = \gcd(p_4, g_X).$$

Over a splitting field define

$$a(d) = \min\{4, \text{ord}_E r_d\} \in \{0, 1, 2, 3, 4\},$$

and

$$b_i(d) = \min\{2, \text{ord}_{Q_i} r_d\} \in \{0, 1, 2\}.$$

For a nonzero datum the exact contact label $(a, b_1, \dots, b_5)$ is realizable if and only if

$$a + \sum_{i=1}^5 b_i \leq 7.$$

On this stratum,

$$g_X = E^a \prod_i Q_i^{b_i},$$

while g_F and g_H are obtained by truncating exponents to $(3, 1)$ and $(1, 1)$ respectively. The stratum has

$$\dim = 8 - a - \sum_i b_i.$$

There are 708 nonzero fine strata and one zero stratum. The canonical full-lift annihilator is

$$\mu_X^{\text{can}} = \frac{p_5}{g_X} = E^{4-a} \prod_i Q_i^{2-b_i}.$$

Proof. The divisibility chain follows from $m_A \mid p_4 \mid p_5$. Over the splitting field, gcds are determined by truncated contact orders at the six pairwise distinct spectral points. A nonzero degree- < 8 polynomial divisible by $E^a \prod_i Q_i^{b_i}$ exists exactly when the total contact weight is at most seven; sufficiency is realized by that product itself. Hermite interpolation gives independence of the corresponding jet conditions and hence the dimension formula. Summing feasible labels gives

$$1 + 6 + 21 + 51 + 96 + 146 + 186 + 201 = 708.$$

The zero datum forms the unique final stratum. □

Scheme-theoretically, the three levels have multiplicity profiles

$$\text{Res}(m_A, r_d) : (E, Q_i) = (1, 1),$$

$$\text{Res}(p_4, r_d) : (E, Q_i) = (3, 1),$$

$$\text{Res}(p_5, r_d) : (E, Q_i) = (4, 2).$$

The third level therefore adds precisely the fourth resonant jet and second-order contact with each quintic spectral hyperplane.

Over $\mathbb{Q}$, irreducibility of q and the degree bound $\deg r_d < 8$ imply that the nonzero possibilities for g_X are exactly

$$1, E, E^2, E^3, E^4, q, Eq, E^2q.$$

Thus the rational canonical three-level stratification consists of eight nonzero strata plus the zero datum.

Remark 17.5 (Scope). *Theorem 17.4 classifies the canonical polynomial lift in the fixed cubic x_5 module. It is not a claim that every inverse-Leibniz contraction has the same multiplicity pattern. The general problem is to determine the analogue of the exponent truncation rules from the spectral multiplicity vectors of the homogeneous operator and the triangular forcing maps.*

The corresponding structures are illustrated in Figure 14.

Fix the canonical polynomial datum r_d of degree < 8 in $\mathcal{D}_5 = \mathbb{C}[S]/(p_4)$.

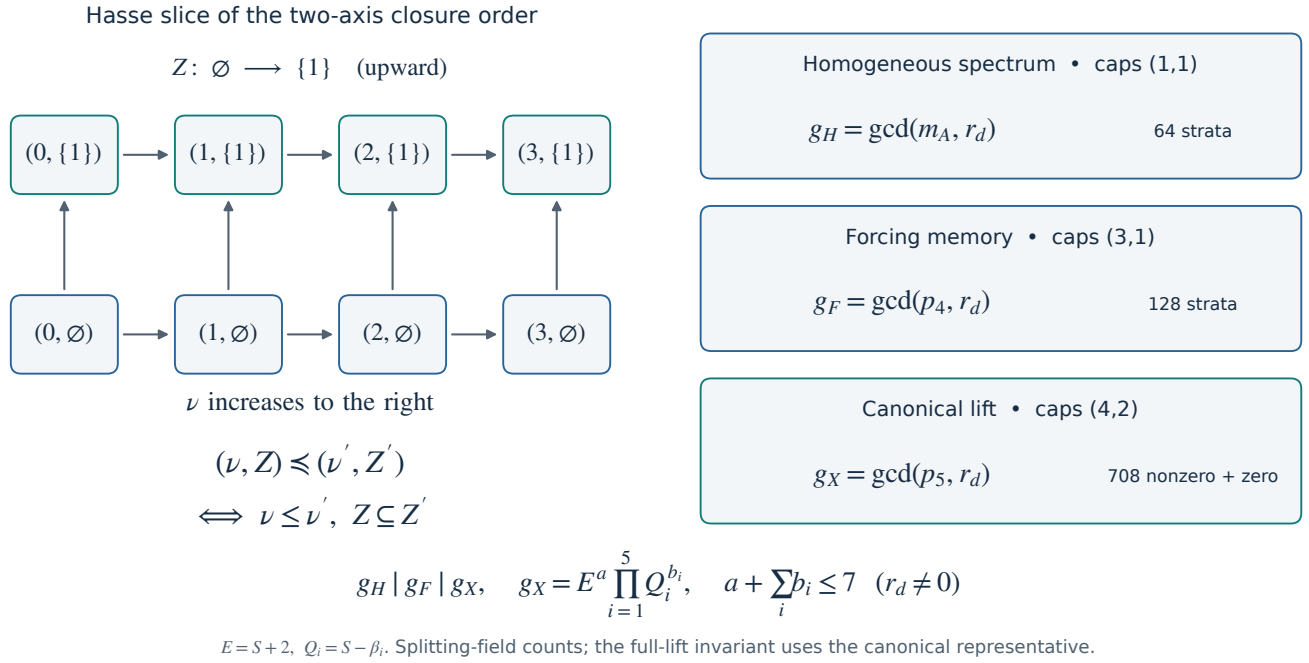


FIGURE 14. Three levels of spectral contact. The left diagram is an eight-vertex slice of the two-axis closure order, with all other quintic roots excluded from Z ; arrows point toward additional vanishing. The full splitting-field counts are 64, 128, and 708 nonzero fine strata plus the zero datum. The third level uses the specified degree- < 8 polynomial lift, not an $\mathbb{C}[S]$ -linear splitting.

18. CONCLUSION AND FIXED-CASE SCOPE

For the fixed cubic data studied here, the reconstruction calculation and its chosen-contraction spectral analysis give the following results:

- (i) the weak affine reconstruction fiber, strict bimodule equations, tangent observability, and explicit global non-identifiability;
- (ii) the formal germ $\mathbb{C}[[u, v]]/(v^3)$, the dual-number and cubic obstruction calculations, one complete contraction with post-cubic vanishing through arity 23, and a second complete contraction with a nonzero frozen arity-23 operation;
- (iii) the full degree-two splitting with $\dim H^2 = 42$, and the autonomous low-rail results with frozen OOS certificates at Y_{17} and Y_{18} ;
- (iv) the exact residual-module, landing-operator, and three-level spectral stratifications, giving 708 nonzero splitting-field strata plus the zero datum and nine rational strata in total.

The new feedback theorem includes all entries, right-hand sides, solutions, and pivot inverses. Historical detector sensitivities and quotient actuator flags remain unrecovered and are excluded from the proof chain. In the spanning examples the pointwise fixed- B stabilizer is

trivial, so the four-point ambiguity is genuine. The unified companion supplies the new matrix certificates, the earlier frozen comparison, and the separate low-rail and companion-paper verification chain.

Paper III [13] gives a finite normalized-tilt spectral-floor theorem for marked data. A classification of arbitrary triangular forcing and its exponent-truncation rules remains distinct from that result and from the fixed calculation here; see Section 19.

19. OPEN PROBLEMS AND NEXT STEPS

Problem 19.1 (Relative presentation versus intrinsic normal forms). *Theorem 12.12 gives a nonzero presentation quartic class, while Theorem 12.19 shows that it disappears after restoring the full Hochschild differential. Construct a relative theory, preferably in terms of the mapping cone or homotopy fiber of*

$$\Psi : \mathfrak{g}_{B,p} \rightarrow \mathfrak{p}_{B,p},$$

that measures exactly which presentation normal-form obstructions are truncation artifacts and which survive intrinsically.

Problem 19.2 (General triangular spectral-extension theory). *The fixed cubic contraction now admits an all-orders low-rail theorem and complete landing stratifications. Abstract these results to triangular systems whose homogeneous operator has spectral multiplicity data*

$$m_A(S) = \prod_{\lambda} (S - \lambda)^{e_{\lambda}}$$

and whose forcing branches are built from linear maps and Cauchy convolutions of lower rails. Determine general conditions under which forcing annihilators, residual landing modules, and higher-lift contact are obtained by deterministic exponent-truncation rules analogous to Theorems 15.4 and 17.4.

Problem 19.3 (Beyond finite feedback). *Determine whether one contraction can retain the cubic operation and make every higher degree-one operation vanish, or whether a later finite obstruction persists for every admissible completion. The two arity-23 outcomes above do not settle this question. No future actuator flag is assumed.*

These three questions are separate from the fixed results proved here. Broader gauge quotients, graded reconstruction and geometric applications require additional input and are not part of the present closeout.

APPENDIX A. EXACT COMPUTATIONAL VERIFICATION

The portable v0.08 companion has three evidence directories: **legacy** (the preserved v0.04 finite suites), **transfer_v05** (the frozen full-completion comparison), and **feedback_v06** (the compatible feedback contraction). The root command `python -B verify_all.py` runs all three evidence families and the old-trajectory incompatibility countercheck. The companion is **three_papers_v0_08_bundle.zip**. Its **EVIDENCE_MAP.md** identifies each claim's exact file and replay entry point. No public archive identifier has yet been assigned. Use Python with assertions enabled and SymPy 1.14.0. No unpublished helper outside the package is needed for these commands.

Foundations and the formal germ. The legacy foundation and recovery scripts reconstruct the weak-action matrices from the specified algebra products and fixed B , substitute the strict equations, and compute exact Groebner bases. They check the four-point, dual-number, square-zero and cubic examples, including the cubic formal germ, the rational higher-tail certificate and the finite-flat multiplication matrices. They also reconstruct the presentation quartic obstruction and the intrinsic sources through arity seven, checking the displayed degree-three minors. The relation morphism and general obstruction arguments are proved in the body; finite examples are checks, not substitutes for those proofs.

The unrestricted presentation certificate. The root **verify_body_corrections.py** reconstructs the weak nine-dimensional subspace and all 54 unit-preserving axis coordinates. It verifies every identity for the six-coordinate functional $\mathcal{J}$, including arbitrary symbolic tangent parameters. Its relaxed coupled quartic matrix has ranks 57/58 before and after adjoining the forcing column. A separate exact example cancels the single uv^3 source, preventing reuse of the former overstrong single-coefficient assertion. The script also checks the two-window temporal ranks needed for the landing quotient definitions.

Two full contractions, kept distinct. The **transfer_v05** consumer rebuilds the full d_2 kernel and the projection from the saved exact inverse, verifies all sources through arity 22, then evaluates the next source without changing the projection. Its regenerated complete-splitting and held-out files must match the supplied files byte for byte.

The **feedback_v06** consumer uses a separate flattened cup-product implementation. It recomputes the full 30375-by-2025 differential, checks $Z^2 = B^2 \oplus i_2 H^2$, the full columns of h_+, p_+ , all 100 saved assignments, every quadratic pair and every source through arity 23. For each of the five matrices it checks the block formula, CSV/JSON equality, source-derived right-hand side, rational solution, pivot inverse and determinant. All arithmetic is exact over $\mathbb{Q}$. The metadata-only ranks in the older v0.32 file are not used as a substitute for these new matrix checks.

Coordinates and file linkage. Normalized algebra inputs are the 15 lexicographically ordered pairs in $\{0, 1, 2, 3\}^2 \setminus \{(0, 0)\}$. If $\text{sid}(s)$ is the zero-based input index and $0 \leq r, c < 3$, the ambient one-cochain coordinate is $9 \text{sid}(s) + 3r + c$. The two-cochain coordinate is $9(15 \text{sid}(s) + \text{sid}(t)) + 3r + c$. Sparse rational vectors use triples [index, numerator, denominator]. Polynomial coefficients use $[a, b]$ for $u^a v^b$. The 135 ambient one-cochain coordinates contain the 90-dimensional constrained space and its 88-dimensional acyclic complement. The two new harmonic sector coordinates embed into the full 42-dimensional cohomology via the matrix in **blocks.json**. The files **projection.json**, **homotopy.json**, and **transfer.json** specify the complete maps and recursion. The nonzero 4/9 comparison witness belongs only to **transfer_v05/heldout_arity23.json**.

Low rails and spectral selection. The seven legacy fixed-case verifiers check the v0.32/v0.33 frozen low-rail predictions, the cyclic matrix A_* , the residual modules of ranks (1, 2, 8), the

exact landing matrices, gcd truncations and the 64, 128 and 709 finite stratum counts. The all-orders assertions follow from the operator, forcing and finite-rank arguments in Sections 15–17; two out-of-sample identities are separate finite evidence. The polynomial and module claims are relative to the stated fixed low-rail data, not to every contraction or arbitrary triangular forcing system.

Verification boundary. The original historical detector pair, distinguished sensitivity, specific pivot minors and evolving quotient-history flags have not been recovered. Their old statements remain in the preserved research archive, not in this manuscript's theorem chain. Computational replay is an exact rational certificate check, not a proof-assistant formalization of the complete papers. The bibliography gives a bounded primary-literature positioning, not an exhaustive priority search. No result depends on a prospective research illustration.

APPENDIX B. LOW-ORDER SOURCE COEFFICIENTS AND JOINT CERTIFICATES

This appendix supplies the explicit data summarized in Table 2. The constructions retain every earlier splitting assignment and use the joint-source placement criterion.

B.1. The intrinsic contraction extends through arity five. The arity-four theorem leaves genuine freedom in the values of the contracting homotopy on the two nonclosed quartic source cochains. To continue one *single* contraction, this freedom must be chosen compatibly with both the already selected quartic acyclic directions and the new quintic sources. We now give an explicit compatible choice.

Let E_{ij} again denote the standard matrix units. Define constrained one-cochains $\mathbf{p}, \mathbf{q} \in \mathfrak{g}_{B,p}^1$, with zero mixed values, by

$$\begin{aligned} X_{\mathbf{p}}(\epsilon) &= -\frac{1}{3}E_{21} + \frac{1}{6}E_{22}, & Y_{\mathbf{p}}(\epsilon) &= \frac{1}{3}E_{21} - \frac{1}{6}E_{22}, \\ X_{\mathbf{p}}(\epsilon^2) &= -\frac{2}{3}E_{22}, & Y_{\mathbf{p}}(\epsilon^2) &= \frac{2}{3}E_{22}, \\ X_{\mathbf{p}}(\epsilon^3) &= -E_{23}, & Y_{\mathbf{p}}(\epsilon^3) &= E_{23}, \end{aligned}$$

and

$$\begin{aligned} X_{\mathbf{q}}(\epsilon) &= -\frac{1}{2}E_{22}, & Y_{\mathbf{q}}(\epsilon) &= \frac{1}{2}E_{22}, \\ X_{\mathbf{q}}(\epsilon^2) &= -E_{23}, & Y_{\mathbf{q}}(\epsilon^2) &= E_{23}, \\ X_{\mathbf{q}}(\epsilon^3) &= 0, & Y_{\mathbf{q}}(\epsilon^3) &= 0. \end{aligned}$$

Exact calculation gives

$$(37) \quad \text{rank span}\{de, dk, d\mathbf{p}, d\mathbf{q}\} = 4,$$

so the degree-one acyclic complement may be chosen to contain all four cochains simultaneously.

Set

$$(38) \quad C_{13} = Y_{13} - d\mathbf{p}, \quad C_{04} = Y_{04} - d\mathbf{q}.$$

Then the quartic source decomposition is compatible with

$$h(Y_{13}) = \mathbf{p}, \quad h(Y_{04}) = \mathbf{q},$$

while C_{13} and C_{04} are placed in the degree-two acyclic complement. Since $Y_{22} = 4x \smile x$ and $h(x \smile x) = -e$, the arity-four Taylor coefficient of the transferred section can be chosen as

$$(39) \quad F_4 = 4u^2v^2e - uv^3\mathbf{p} - v^4\mathbf{q}.$$

The quintic Maurer–Cartan source

$$R_5 = [F_1, F_4] + [F_2, F_3]$$

has only the four monomial coefficients

$$(40) \quad Z_{32} = -8x \smile x,$$

$$(41) \quad Z_{23} = 4s - [a, p],$$

$$(42) \quad Z_{14} = -[a, q] - [x, p] - 4e \smile e,$$

$$(43) \quad Z_{05} = -[x, q] + [e, k].$$

Here Z_{32} is a boundary because

$$Z_{32} = d(8e).$$

The remaining question is whether the nonclosed quartic complement directions and the three non-boundary-type quintic sources can be put into the *same* acyclic complement.

Theorem B.1 (Intrinsic quintic vanishing extension). *For the cubic reconstruction of Section 12, the intrinsic contraction of Theorem 12.19 can be chosen and extended so that*

$$\boxed{\ell_4|_{\text{Sym}^4 H^1} = 0, \quad \ell_5|_{\text{Sym}^5 H^1} = 0.}$$

The cubic obstruction $[\hat{\zeta}]$ remains nonzero.

Proof. Let

$$W_{4,5}^{\text{nc}} = \text{span}\{C_{13}, C_{04}, Z_{23}, Z_{14}, Z_{05}\} \subset C^2(S, E_\mu).$$

We show that the Hochschild differential is injective on this five-dimensional space.

Write

$$x = \epsilon \otimes 1, \quad y = 1 \otimes \epsilon^{\text{op}}, \quad m = \epsilon \otimes \epsilon^{\text{op}}.$$

For a normalized three-cochain Φ , evaluate successively the coordinates

$$\Phi(y, y, y)_{12}, \quad \Phi(y, y, y)_{13}, \quad \Phi(y, y, y)_{23}, \quad \Phi(y, y, m)_{13}, \quad \Phi(y, x, y)_{13}.$$

On the ordered columns

$$dC_{13}, dC_{04}, dZ_{23}, dZ_{14}, dZ_{05},$$

these five functionals give the exact matrix

$$(44) \quad \begin{pmatrix} 0 & -\frac{1}{2} & 0 & -\frac{2}{3} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{3} & \frac{1}{3} & -1 \\ 0 & 1 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{3} & -\frac{1}{3} & 0 \end{pmatrix}, \quad \det = \frac{1}{18} \neq 0.$$

Hence

$$\text{rank span}\{dC_{13}, dC_{04}, dZ_{23}, dZ_{14}, dZ_{05}\} = 5,$$

and therefore

$$W_{4,5}^{\text{nc}} \cap Z^2 = 0.$$

The degree-two acyclic complement can consequently be chosen to contain the whole space $W_{4,5}^{\text{nc}}$ at once. Together with the boundary Z_{32} , every quintic source coefficient has zero cohomology projection. The same splitting already kills all quartic coefficients by construction, so the resulting single contraction satisfies

$$\ell_4|_{\text{Sym}^4 H^1} = \ell_5|_{\text{Sym}^5 H^1} = 0.$$

Nothing in this extension changes the cubic representative $\hat{\zeta}$, whose nonvanishing was established by (23). $\square$

Remark B.2 (No arity-five obstruction for this intrinsic splitting). *The arity-five calculation is stronger than checking the quintic sources in isolation. The determinant (44) tests the joint span of the previously fixed quartic acyclic directions and the new quintic directions. Thus the argument really constructs one compatible contraction through arity five; it does not silently replace the contraction between orders.*

B.2. The intrinsic contraction extends through arity six. We now continue the *same* contraction one order further. The arity-five theorem fixed the degree-two complement directions but left freedom in the values of the homotopy on the three non-boundary-type quintic sources. We choose explicit constrained one-cochains

$$\mathbf{r}, \mathbf{w}, \mathbf{t} \in \mathfrak{g}_{B,p}^1$$

with the following values. The cochain $\mathbf{r}$ has only factor-axis values

$$\begin{aligned} X_{\mathbf{r}}(\epsilon) &= -\frac{1}{3}E_{31} + \frac{1}{6}E_{32}, & Y_{\mathbf{r}}(\epsilon) &= \frac{1}{3}E_{31} - \frac{1}{6}E_{32}, \\ X_{\mathbf{r}}(\epsilon^2) &= -\frac{2}{3}E_{32}, & Y_{\mathbf{r}}(\epsilon^2) &= \frac{2}{3}E_{32}, \\ X_{\mathbf{r}}(\epsilon^3) &= -E_{33}, & Y_{\mathbf{r}}(\epsilon^3) &= E_{33}. \end{aligned}$$

The cochains $\mathbf{w}$ and $\mathbf{t}$ have zero factor-axis values and are supported only at

$$m = \epsilon \otimes \epsilon^{\text{op}},$$

where

$$\mathbf{w}(m) = E_{11}, \quad \mathbf{t}(m) = E_{12}.$$

Exact calculation gives

$$(45) \quad \text{rank span}\{d\mathbf{e}, d\mathbf{k}, d\mathbf{p}, d\mathbf{q}, d\mathbf{r}, d\mathbf{w}, d\mathbf{t}\} = 7.$$

Thus all seven cochains may be included simultaneously in one degree-one acyclic complement.

Set

$$(46) \quad D_{23} = Z_{23} - d\mathbf{r}, \quad D_{14} = Z_{14} - d\mathbf{w}, \quad D_{05} = Z_{05} - d\mathbf{t}.$$

This corresponds to choosing

$$h(Z_{23}) = \mathbf{r}, \quad h(Z_{14}) = \mathbf{w}, \quad h(Z_{05}) = \mathbf{t}.$$

Since $Z_{32} = d(8\mathbf{e})$, the arity-five Taylor coefficient can then be chosen as

$$(47) \quad F_5 = -8u^3v^2\mathbf{e} - u^2v^3\mathbf{r} - uv^4\mathbf{w} - v^5\mathbf{t}.$$

The sextic Maurer–Cartan source is

$$R_6 = [F_1, F_5] + [F_2, F_4] + \frac{1}{2}[F_3, F_3].$$

It has five monomial coefficients:

$$(48) \quad U_{42} = -8[\mathbf{a}, \mathbf{e}],$$

$$(49) \quad U_{33} = -[\mathbf{a}, \mathbf{r}] - 8\mathbf{s},$$

$$(50) \quad U_{24} = -[\mathbf{a}, \mathbf{w}] - [\mathbf{x}, \mathbf{r}] + 12\mathbf{e} \smile \mathbf{e},$$

$$(51) \quad U_{15} = -[\mathbf{a}, \mathbf{t}] - [\mathbf{x}, \mathbf{w}] - [\mathbf{e}, \mathbf{p}] - 2[\mathbf{e}, \mathbf{k}],$$

$$(52) \quad U_{06} = -[\mathbf{x}, \mathbf{t}] - [\mathbf{e}, \mathbf{q}] + \mathbf{k} \smile \mathbf{k}.$$

Here $\mathbf{s} = [\mathbf{x}, \mathbf{e}]$ is the nonzero cubic source fixed in the intrinsic construction above. The coefficient U_{42} is a boundary. Indeed,

$$[\mathbf{a}, \mathbf{e}] = -2\mathbf{x} \smile \mathbf{x}, \quad d\mathbf{e} = -\mathbf{x} \smile \mathbf{x},$$

so

$$(53) \quad U_{42} = d(-16\mathbf{e}).$$

Theorem B.3 (Intrinsic sextic vanishing extension). *For the cubic reconstruction of Section 12, one intrinsic contraction can be chosen so that*

$$\boxed{\ell_4|_{\text{Sym}^4 H^1} = \ell_5|_{\text{Sym}^5 H^1} = \ell_6|_{\text{Sym}^6 H^1} = 0.}$$

The cubic obstruction $[\hat{\zeta}]$ remains nonzero.

Proof. Let

$$W_{4,5,6}^{\text{nc}} = \text{span}\{C_{13}, C_{04}, D_{23}, D_{14}, D_{05}, U_{33}, U_{24}, U_{15}, U_{06}\}.$$

We prove that the Hochschild differential is injective on this nine-dimensional space. As before set

$$x = \epsilon \otimes 1, \quad y = 1 \otimes \epsilon^{\text{op}}, \quad m = \epsilon \otimes \epsilon^{\text{op}}.$$

Evaluate degree-three cochains successively at the nine coordinates

$$(yyy)_{11}, (yyy)_{12}, (yyy)_{13}, (yyy)_{23}, (yyy)_{32}, (yym)_{13}, (yym)_{23}, (yy^2y)_{13}, (yxy)_{13}.$$

On the ordered columns

$$dC_{13}, dC_{04}, dD_{23}, dD_{14}, dD_{05}, dU_{33}, dU_{24}, dU_{15}, dU_{06},$$

these functionals give the exact matrix

$$(54) \quad \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & -\frac{2}{3} & -\frac{1}{2} & -\frac{1}{3} & -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{3} & \frac{1}{3} & -1 & -\frac{1}{6} & \frac{1}{6} & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{3} & 0 & -\frac{1}{3} & \frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 0 & -1 & -1 & -\frac{1}{3} & -\frac{2}{3} & 2 & 1 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{3} & -\frac{1}{3} & 0 & \frac{1}{6} & \frac{5}{6} & 0 & 1 \end{pmatrix}, \quad \det = -\frac{1}{486} \neq 0.$$

Hence

$$\text{rank span}\{dC_{13}, dC_{04}, dD_{23}, dD_{14}, dD_{05}, dU_{33}, dU_{24}, dU_{15}, dU_{06}\} = 9,$$

and therefore

$$W_{4,5,6}^{\text{nc}} \cap Z^2 = 0.$$

The degree-two acyclic complement can consequently be chosen to contain all nine directions simultaneously. Together with the boundary coefficient U_{42} , every sextic source coefficient has zero cohomology projection. The same degree-one and degree-two splittings already realize the quartic and quintic vanishings, so this single contraction satisfies the claimed identities through arity six. None of these choices alters the previously fixed nonzero cubic class $[\hat{\zeta}]$. $\square$

Remark B.4 (No intrinsic obstruction through arity six). *The arity-six test again uses the joint source space rather than the sextic coefficients in isolation. Thus Theorem B.3 constructs one compatible contraction through arity six. It does not prove that all higher transferred operations vanish. Arity seven is the next point at which a new closed non-boundary combination may appear inside the growing joint acyclic source space.*

B.3. The intrinsic contraction extends through arity seven. We keep every splitting choice from the arity-six construction and choose homotopy lifts for the four nonclosed sextic coefficients. Let

$$m = \epsilon \otimes \epsilon^{\text{op}}.$$

Define four constrained one-cochains A, B, C, D with zero factor-axis values, supported only at m , by

$$A(m) = E_{21}, \quad B(m) = E_{22}, \quad C(m) = E_{23}, \quad D(m) = E_{31}.$$

Exact calculation gives

$$(55) \quad \text{rank span}\{de, dk, dp, dq, dr, dw, dt, dA, dB, dC, dD\} = 11.$$

Thus all eleven degree-one homotopy directions can be included simultaneously in one acyclic splitting.

Set

$$(56) \quad G_{33} = U_{33} - dA, \quad G_{24} = U_{24} - dB, \quad G_{15} = U_{15} - dC, \quad G_{06} = U_{06} - dD.$$

Accordingly the arity-six Taylor coefficient can be chosen as

$$(57) \quad F_6 = 16u^4v^2e - u^3v^3A - u^2v^4B - uv^5C - v^6D.$$

The arity-seven Maurer–Cartan source is

$$R_7 = [F_1, F_6] + [F_2, F_5] + [F_3, F_4].$$

Its six monomial coefficients are

$$(58) \quad V_{52} = 16[a, e],$$

$$(59) \quad V_{43} = -[a, A] + 16[x, e],$$

$$(60) \quad V_{34} = -[a, B] - [x, A] - 16[e, e],$$

$$(61) \quad V_{25} = -[a, C] - [x, B] - [e, r] + 2[e, p] + 4[k, e],$$

$$(62) \quad V_{16} = -[a, D] - [x, C] - [e, w] + 2[e, q] - [k, p],$$

$$(63) \quad V_{07} = -[x, D] - [e, t] - [k, q].$$

The leading coefficient is again a boundary. Since $[a, e] = -2x \smile x$ and $de = -x \smile x$,

$$(64) \quad V_{52} = d(32e).$$

Theorem B.5 (Intrinsic arity-seven vanishing extension). *For the cubic reconstruction of Section 12, one intrinsic contraction can be chosen so that*

$$\boxed{\ell_4|_{\text{Sym}^4 H^1} = \ell_5|_{\text{Sym}^5 H^1} = \ell_6|_{\text{Sym}^6 H^1} = \ell_7|_{\text{Sym}^7 H^1} = 0.}$$

The cubic obstruction $[\hat{\zeta}]$ remains nonzero.

Proof. Use the actual sextic complement vectors from (56) and define

$$W_{4,5,6,7}^{\text{nc}} = \text{span}\{C_{13}, C_{04}, D_{23}, D_{14}, D_{05}, G_{33}, G_{24}, G_{15}, G_{06}, V_{43}, V_{34}, V_{25}, V_{16}, V_{07}\}.$$

Because $d^2 = 0$, the differentials of $G_{33}, G_{24}, G_{15}, G_{06}$ are the same as those of $U_{33}, U_{24}, U_{15}, U_{06}$ used in the arity-six certificate. It therefore suffices to prove that the five new degree-seven directions increase the joint differential rank from nine to fourteen.

As before write

$$x = \epsilon \otimes 1, \quad y = 1 \otimes \epsilon^{\text{op}}, \quad m = \epsilon \otimes \epsilon^{\text{op}}.$$

On the ordered columns

$$dC_{13}, dC_{04}, dD_{23}, dD_{14}, dD_{05}, dG_{33}, dG_{24}, dG_{15}, dG_{06}, dV_{43}, dV_{34}, dV_{25}, dV_{16}, dV_{07},$$

consider the fourteen coordinate evaluations

$$(yyy)_{11}, (yyy)_{12}, (yyy)_{13}, (yyy)_{21}, (yyy)_{22}, (yyy)_{23}, (yyy)_{32}, \\ (yyx)_{11}, (yyx)_{12}, (yym)_{13}, (yym)_{23}, (yy^2y)_{13}, (yxy)_{13}, (yxy)_{23}.$$

The resulting exact 14×14 minor, denoted M_7 , satisfies

$$(65) \quad \det M_7 = \frac{11}{34992} \neq 0.$$

Hence

$$\text{rank } d|_{W_{4,5,6,7}^{\text{nc}}} = 14, \quad W_{4,5,6,7}^{\text{nc}} \cap Z^2 = 0.$$

Thus the degree-two acyclic complement can be enlarged to contain all fourteen directions simultaneously. Together with the boundary coefficient V_{52} , every arity-seven source coefficient has zero cohomology projection. Equation (55) guarantees that the required eleven degree-one homotopy directions coexist in the same splitting. Therefore the same intrinsic contraction realizes the quartic, quintic, sextic, and arity-seven vanishings while preserving the previously fixed nonzero cubic class. $\square$

Remark B.6 (No intrinsic obstruction through arity seven). *The rank-fourteen certificate is a simultaneous history test: it includes every nonclosed direction already committed at arities four, five, and six. Thus one contraction realizes the arity-seven vanishing. This argument alone leaves arity eight unchecked; Theorem 13.1 extends the verified range without asserting an all-orders pure-cubic transfer.*

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